

Bias-in-debias-out: Hierarchical channel-spatial bias calibration for cross-domain few-shot classification

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ABSTRACT

The core challenge of cross-domain few-shot learning (CD-FSL) stems from models' inability to generalize source-domain inductive biases to target domains under significant distribution shifts. While existing methods predominantly employ strategies like auxiliary target data adaptation, feature disentanglement, or metric space alignment, they overlook two inherent biases entrenched during source-domain training: (1) channel-wise dependency on source-specific feature patterns and (2) spatial-wise preference for source-typical structures, both of which hinder cross-domain transfer. We propose the first unified Channel-Spatial Dual-dimensional Bias Calibration (CSDBC) framework to systematically address these biases through progressive dilution, recomposition, and alignment. Our approach integrates three key innovations: (1) a parameter-free Static Base-class Bias Dilution (SBBD) module that dilutes source-specific channel-spatial biases through layer-wise and point-wise modulation, effectively suppressing overfitting to source-specific patterns; (2) a Dynamic Novel-class Bias Recomposition (DNBR) module that generates target-adaptive channel-spatial soft masks via meta-optimized lightweight depth-wise separable convolutions, enabling target-domain channel reweighting and spatial preference adjustment; and (3) a Novel-class Cross-image Semantic Alignment (NCSA) module that establishes channel correlations and spatial correspondences between support-query pairs, significantly enhancing both discriminability and semantic consistency of target-domain features. Extensive experiments across eight CD-FSL benchmarks demonstrate consistent improvements, outperforming SOTA methods by 1.35 % (5-way 1-shot) and 2.00 % (5-way 5-shot) in average accuracy under varying domain shifts.

1. Introduction

This work tackles the practical challenge of cross-domain few-shot learning (CD-FSL). While existing few-shot models can rapidly acquire novel-class recognition from limited samples [1–3], real-world scenarios often exhibit significant domain shifts between base and novel classes [4–6]. CD-FSL poses greater challenges than standard domain adaptation: the extreme scarcity of novel-class samples (typically 1–5 per class) provides insufficient data to bridge the domain gap, severely impairing recognition accuracy.

In standard few-shot learning (FSL), base and novel classes share the same domain but have distinct category labels, with each novel class containing only a few labeled samples. Researchers have effectively learned cross-task inductive biases [7] and reasoning mechanisms [8,9] by deliberately constructing n -way k -shot classification scenarios during training and testing. While mainstream FSL methods [7,9–11]—primarily optimized through metric learning and meta-learning—excel at in-domain novel class recognition, their performance degrades significantly under domain shifts, with some complex mod-

els underperforming even simple fine-tuning approaches [5,12]. Though domain adaptation (DA) methods [13–17] improve cross-domain generalization via adversarial training of domain-invariant features, they typically require abundant unlabeled target-domain data with consistent category labels. This assumption becomes invalid in CD-FSL, where base and novel classes are disjoint and target samples are extremely scarce, posing unique challenges that existing methods fail to address.

Fundamentally, CD-FSL simultaneously faces the dual challenges of FSL and DA, making simple combinations of existing methods ineffective. Recent approaches address domain shifts in few-shot settings through two primary strategies. The first category [6,18,19] constructs sophisticated training scenarios to prevent overfitting to domain-specific knowledge—for instance, DSL [18] employs “domain-switch learning” to rapidly alternate training domains with constraints, simulating cross-domain n -way k -shot conditions, while ATA [19] develops adversarial task augmentation to create challenging virtual tasks that enhance robustness. The second category [6,20–23] directly incorporates DA techniques via adversarial learning and feature transformations, such as

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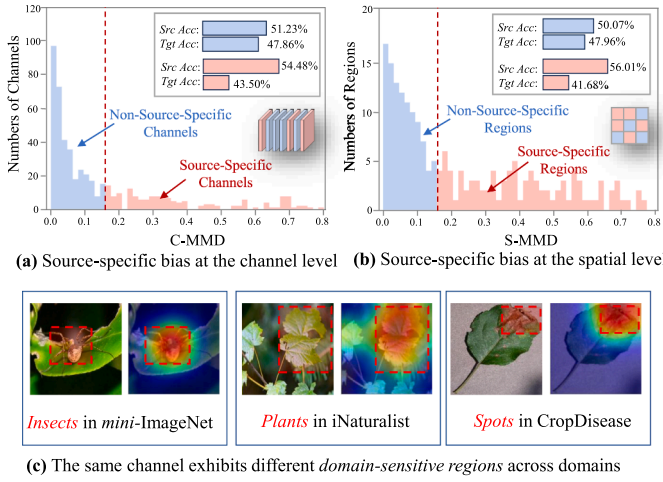


Fig. 1. Cross-domain bias analysis via C-MMD and S-MMD. (a) C-MMD shows source-specific channels (red) suffer target-domain accuracy drops (54.48% → 43.50%), while domain-invariant channels (blue) maintain stable performance. (b) S-MMD reveals similar spatial-level bias, with red regions exhibiting domain sensitivity. (c) Attention heatmaps confirm divergent activation patterns—the same channel focuses on insect textures (*mini-ImageNet*), leaf shapes (*iNaturalist*), or disease spots (*CropDisease*), proving cross-domain bias in spatial preference. (C-MMD: Channel-level Maximum Mean Discrepancy; S-MMD: Spatial-level Maximum Mean Discrepancy; Features are extracted from the penultimate layer of ResNet-10, with *mini-ImageNet* as the source dataset and eight cross-domain datasets as the target datasets).

LFT [4]’s learnable feature transform layers that simulate cross-domain feature distributions, and AFA [22]’s adversarial feature augmentation for target-domain distribution alignment. Though effective, these methods often require stringent training conditions and complex procedures, frequently necessitating retraining for new tasks or suffering from overfitting. Transductive-based approaches [24–26] leverage auxiliary target-domain datasets with limited labeled samples, yet the significant source-target data imbalance typically causes source overfitting, where models overemphasize source-specific features at the expense of target-domain generalization.

This work pioneers a systematic characterization of cross-domain bias using channel-level maximum mean discrepancy (C-MMD) [27] and spatial-level maximum mean discrepancy (S-MMD), revealing the bias mechanisms inherent in both channel and spatial dimensions of features. Specifically, C-MMD quantifies channel-level source overfitting by measuring distributional divergence between source and target domains along channel dimensions based on maximum mean discrepancy theory, while S-MMD evaluates analogous spatial-wise discrepancies. Experimental analyses demonstrate that: (1) Channel-wise (Fig. 1(a)), low C-MMD channels (blue zones) maintain consistent classification performance across both source (51.23%) and target (47.86%) domains, whereas high C-MMD channels (red zones) show severe target-domain degradation (54.48% → 43.50%) despite superior source-domain performance; (2) Spatial-wise (Fig. 1(b)), low S-MMD regions exhibit cross-domain consistency contrasting with significant performance drops in high S-MMD areas; (3) Fig. 1(c) provides further visual evidence of divergent spatial activation patterns: for similar image representations, the same channel exhibits completely different region preferences across domains (e.g., insect-focused activations in *mini-ImageNet* vs. plant-leaf responses in *iNaturalist* vs. disease-spot concentrations in *CropDisease*). These findings conclusively validate the dual-bias hypothesis: contemporary models simultaneously suffer from channel-wise dependency on source-specific feature patterns and spatial-wise preference for source-domain structures.

Building upon this discovery, we propose a channel-spatial dual-dimensional bias calibration (CSDBC) framework for CD-FSL. It sys-

tematically suppresses and calibrates inherent source-domain inductive biases via source-domain bias dilution and target-domain bias recombination. Here, “source-domain inductive biases” refer to two intrinsic types: (1) channel-wise bias in feature responses and (2) spatial-wise bias in structural preferences, as illustrated in Fig. 1(upper). The framework operates through three key mechanisms: first, the static base-class bias dilution (SBBB) module employs feature modulation with no learnable parameters to weaken the model’s dependency on source-specific channel responses and spatial structural preferences without significantly compromising its original representational capacity; next, the dynamic novel-class bias recombination (DNBR) module utilizes meta-optimized lightweight depthwise separable convolutions to generate target-adaptive channel importance weights and spatial preference weights, enabling coordinated optimization of dual biases; finally, the novel-class cross-image semantic alignment (NCSA) module establishes bidirectional channel correlations and spatial semantic correspondences between support-query pairs to further enhance target-domain feature discriminability and semantic consistency. Extensive experiments on eight standard cross-domain benchmarks show that our framework significantly outperforms state-of-the-art methods, providing a novel solution for bias calibration in CD-FSL.

Compared with existing CD-FSL methods, our approach exhibits fundamental differences in both problem modeling and bias calibration mechanisms. From the perspective of problem modeling, unlike previous methods that simulate domain shifts through task construction (e.g., DSL [18], ATA [19]) or rely on global feature alignment (e.g., LFT [4], AFA [22]), we, for the first time, systematically characterize dual inductive biases at both the channel and spatial levels from theoretical and empirical perspectives, revealing the channel dependency and spatial activation shift of existing models across different domains. From the perspective of methodological logic, existing methods often rely on additional target domain data [25,26], channel-level cross-domain bias [27–29], or complex training task constructions [18,19], which limits their generalization ability. In contrast, CSDBC offers advantages such as lightweight transferable modulation and task-level semantic alignment, effectively achieving bias suppression and recombination without introducing target domain data. To address the specific characteristics of CD-FSL tasks, we design three innovative mechanisms: (i) parameter-free static base-bias dilution, (ii) lightweight dynamic novel-bias recombination based on sparse meta-learning, and (iii) novel-class cross-image semantic alignment between support and query sets. These modules enable fast and few-shot bias calibration for target domain tasks, collaboratively optimizing feature generalization across both channel and spatial dimensions. Our key contributions are summarized as follows:

- We propose a static base-class bias dilution (SBBB) module that employs parameter-free feature modulation to suppress channel-spatial biases in the source domain, effectively mitigating over-reliance on source-specific features. This work represents the first systematic revelation of dual inherent biases in CD-FSL: channel-wise dependency on source-specific feature patterns and spatial-wise preference for source-domain structures.
- A dynamic novel-class bias recombination (DNBR) module is introduced, which generates target-adaptive channel-spatial soft masks via lightweight depthwise separable convolutions, enabling simultaneous channel reweighting and spatial preference refinement. This is further enhanced by our novel sparse masked-gradient MAML (MG-MAML) strategy for optimized mask parameter updates.
- We develop a novel-class cross-image semantic alignment (NCSA) module that establishes channel correlations and spatial correspondences between support-query pairs, significantly improving target-domain feature discriminability and semantic consistency.
- Collectively, these components form a unified channel-spatial dual-dimensional bias calibration (CSDBC) framework that effectively addresses source-domain inductive biases in CD-FSL. Extensive

experiments under standard CD-FSL settings demonstrate state-of-the-art performance across all benchmarks.

2. Related work

This section reviews recent advances in two pivotal CD-FSL approaches: parameter-tuning strategies that adapt model parameters across domains, and feature-postprocessing methods that refine representations via adaptive transformations.

2.1. Parameter-tuning for CD-FSL

Recent advances in parameter-tuning methods for CD-FSL have coalesced around three principal strategies: parameter freezing, selection, and reweighting. Early meta-learning frameworks like SB-MTL [30] and MPL [31] laid the foundation, with progress toward contemporary approaches like the task-supervised ANIL variant [32], which enhances feature reuse via dynamic weight updates and residual network optimization. Building on this momentum, the parameter selection paradigm has evolved from static architectures to dynamic systems like AugSelect [33]’s subnetwork ensembles and MAP [34]’s modular pipelines, enabling real-time adaptation to target distributions. In parameter reweighting, breakthroughs reveal the dual role of ViT tokens in domain information absorption (CLSToken [35]), while [36] establishes end-to-end adaptation via self-supervised/supervised hybrid learning. These developments are further augmented by style-transfer mechanisms (Wave-SAN [37]) and adversarial feature enhancement (AFA [22]), collectively advancing the field toward fine-grained parameter control. Although parameter-tuning-based CD-FSL methods offer advantages in hypothesis space compression and knowledge transfer efficiency, they have key limitations: the flexibility of parameter freezing and reweighting is restricted by the model architecture; multi-stage parameter adjustments can lead to suboptimal optimization, especially with sparse target-domain data; and these methods are sensitive to source-domain initialization, with high computational overhead in dynamic parameter selection and reweighting, limiting their use in lightweight scenarios.

2.2. Feature-postprocessing for CD-FSL

Recent feature-postprocessing advances for CD-FSL have evolved along three trajectories. Feature selection methods, exemplified by RMFS [38], optimize multi-domain feature pools through dynamic selection mechanisms, while SUR [39] employs parallel feature extractors with adaptive weighting. However, these methods often overlook intrinsic couplings in cross-domain feature interactions. Feature fusion techniques address this limitation through hierarchical integration, as demonstrated by CHEFK [40]’s combination of shallow and deep features, and multi-view frameworks like CDFG [41] that synergize contextual and geometric representations. Nevertheless, their fusion rules remain heuristic and lack theoretical grounding. The most promising developments emerge in feature transformation, where metric-driven approaches (TACDFSL [42]) and module-based methods (FDFSL [43]) have shown particular success. Breakthroughs like DKD-FSL [44] further demonstrate that discriminative knowledge distillation can outperform conventional parameter tuning. Although these methods improve generalization via refined feature optimization, they face challenges: reliance on assumed source-target correlations, high computational cost from multi-layer feature reconstruction and dynamic weighting, and distribution shift risks due to dependence on unlabeled target data—especially when sparse or noisy—requiring better stability and robustness.

Current CD-FSL methods have three key limitations. First, they require complex training procedures and stringent hyperparameter tuning, leading to inefficient training and convergence issues. Second, dynamic feature reconstruction and weight adjustment incur high computational costs, with memory usage growing exponentially for high-

dimensional features. Most importantly, bias correction mainly addresses channel-wise feature biases, neglecting spatial-wise structural biases, limiting models’ cross-domain generalization. To address these limitations, we propose a novel channel-spatial dual-dimensional bias calibration framework that simultaneously resolves channel-wise and spatial-wise biases through a unified approach. Our method combines parameter-free static bias dilution with dynamic meta-optimized adaptation, achieving comprehensive bias calibration efficiently. Extensive experiments demonstrate superior performance across multiple benchmarks, significantly outperforming existing approaches in both accuracy and generalization capability.

3. Proposed method

Formal Problem Definition. CD-FSL addresses the challenge of recognizing previously unseen categories (C^{novel}) in a target domain using minimal training data, by transferring knowledge from a set of base classes (C^{base}) in a source domain. The problem setup requires that C^{novel} and C^{base} are disjoint ($C^{novel} \cap C^{base} = \emptyset$) and that there exists a domain shift between them. During the initial training phase, the model learns from a base dataset $D^{base} = \{(x_i, y_i)\}_{i=1}^N$ where each label y_i belongs to C^{base} . The standard training phase optimizes a cross-entropy objective through feature encoder $g(\cdot)$ and classifier $h(\cdot)$. The pretrained encoder is then adapted to the novel-class dataset D^{novel} containing few-shot examples (typically 1 or 5 per class) with labels from C^{novel} . To ensure rigorous evaluation, approaches [6,18,19] adopt an episodic training and testing paradigm with n -way k -shot tasks.

Overall Framework. This paper proposes a channel-spatial dual-dimensional bias calibration (CSDBC) framework that systematically resolves the dual-bias problem (channel-wise dependency and spatial-wise preference) in CD-FSL through source-domain bias discovery and target-adaptive calibration. As shown in Fig. 2, the framework first incorporates a static base-class bias dilution (SBBD) module, which employs feature modulation with no learnable parameters to weaken the model’s dependency on source-specific channel responses and spatial structure preferences without significantly compromising its original representational capacity. Building upon this, the dynamic novel-class bias recomposition (DNBR) module utilizes lightweight depthwise separable convolutions to generate target-adaptive channel-spatial soft masks, with their parameters optimized through our proposed sparse masked-gradient MAML strategy to ensure proper adjustment of task-critical dimensional masks. Finally, the novel-class cross-image semantic alignment (NCSA) module establishes explicit channel correlations and spatial correspondences between support-query pairs, enhancing discriminative feature learning for the target domain.

Task-Specific Improvements for CD-FSL. To better address the challenges in CD-FSL tasks, we design three task-specific improvement mechanisms: static base-bias dilution (SBBD), dynamic novel-bias recomposition (DNBR), and novel-class cross-image semantic alignment (NCSA). First, to tackle the bias between source and target domains in CD-FSL, SBBD suppresses overly responsive channels and spatial patterns in the source domain, effectively reducing the interference of source domain bias on target tasks without introducing additional parameters. This is particularly suitable for CD-FSL settings where only source domain data are available. Second, considering the scarcity of support set samples in the target domain and the tendency of conventional fine-tuning to overfit, DNBR implements a lightweight and generalizable depthwise separable convolution modulation, combined with a sparse meta-learning strategy, enabling rapid adaptation to the support set and dynamically generating channel/spatial soft masks, thus achieving fast and few-shot bias recomposition for target domain tasks. Finally, to address semantic drift between the support and query sets, NCSA constructs channel and spatial semantic consistency mappings to achieve dual-dimensional alignment between support-query pairs, effectively enhancing matching robustness and semantic consistency in few-shot target domains. These improvement mechanisms, specifically

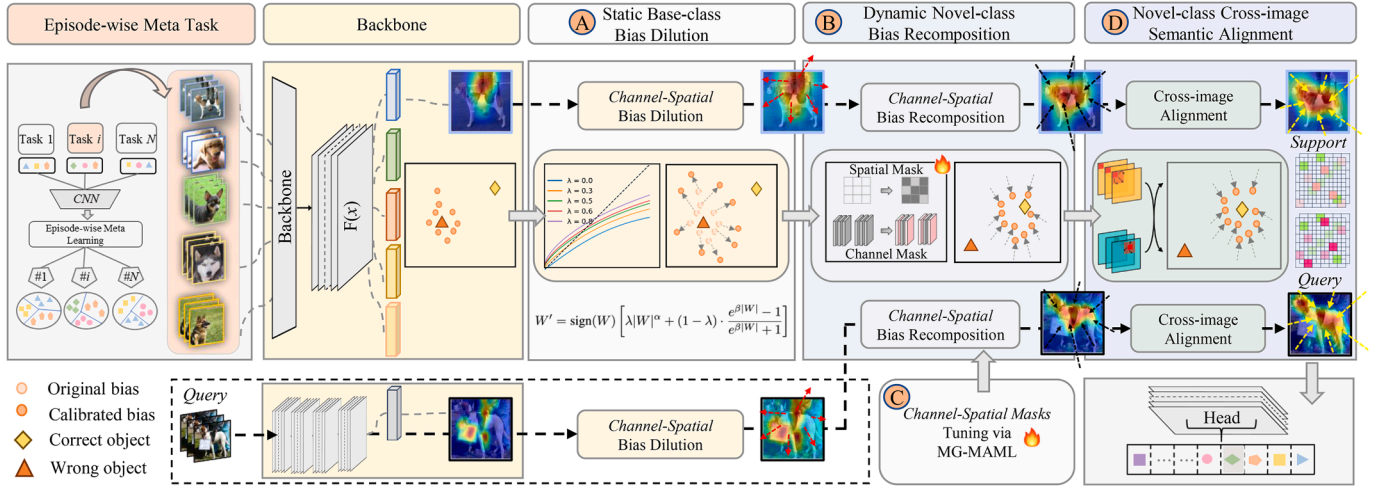


Fig. 2. The proposed CSDBC framework. (A) The SBB module employs feature modulation to suppress source-specific channel responses and spatial preferences (red arrows show bias dilution); (B) the DNBR module generates target-adaptive channel-spatial masks via lightweight depthwise separable convolutions, with black arrows indicating mask-optimized bias recombination; (C) sparse masked-gradient MAML selectively updates critical mask parameters to ensure proper adjustment of task-critical dimensional masks; (D) the NCSA module establishes bidirectional channel correlations and spatial semantic correspondences between support-query pairs to further enhance target-domain feature discriminability and semantic consistency. (yellow arrows show cross-image feature alignment).

designed for CD-FSL, significantly enhance the model's adaptability to the target domain and cross-domain generalization ability.

3.1. Static base-class bias dilution module

Significant domain shifts between training and testing phases often cause discriminative features to deviate from their original distributions. Traditional approaches [6,19,23] typically persist with source-domain knowledge during testing, potentially propagating incorrect inductive biases. Our static bias dilution approach employs a feature modulation function that progressively dilutes channel-spatial biases in the source-domain model through layer-wise and point-wise operations. The key innovation is a parameter-free transformation that simultaneously: (1) compresses over-amplified source-specific activations via power-law suppression, (2) selectively amplifies subtle transferable features using a modified tanh function, and (3) strictly preserves feature signs to maintain latent space geometric consistency.

Given an input image I , the feature representation $\mathbf{z} = g(I) \in \mathbb{R}^m$ is obtained through the feature extractor $g(\cdot)$, out $\{z_i\}_{i=1}^m$ representing individual feature dimensions. The proposed transformation operates on each dimension independently through:

$$\tilde{\phi}_\lambda(z_i) = \text{sign}(z_i) \left[\lambda |z_i|^\alpha + (1 - \lambda) \frac{e^{\beta|z_i|} - 1}{e^{\beta|z_i|} + 1} \right], \quad (1)$$

where the mixing parameter λ governs the trade-off between two complementary operations: a power-law term for head distribution suppression and a modified hyperbolic tangent component for tail distribution enhancement. Default parameter settings use $\alpha = 0.5$ to control suppression strength and $\beta = 1.5$ to adjust nonlinearity characteristics. The complete feature transformation applies this operation across all dimensions:

$$\hat{\phi}_\lambda(\mathbf{z}) = [\tilde{\phi}_\lambda(z_1), \dots, \tilde{\phi}_\lambda(z_m)]. \quad (2)$$

Analysis of the transformation's gradient behavior yields:

$$\frac{\partial \tilde{\phi}_\lambda(z_i)}{\partial z_i} = \lambda \alpha |z_i|^{\alpha-1} + (1 - \lambda) \frac{2\beta e^{\beta|z_i|}}{(e^{\beta|z_i|} + 1)^2}, \quad (3)$$

revealing a dual mechanism where the first component provides controlled gradient reduction for large magnitude activations while the second term preserves substantial gradients for near-zero activations. The sign-preserving architecture ensures consistent handling of both positive

and negative feature values while maintaining numerical stability during optimization. The transformation exhibits two key operational properties: extreme values ($|z_i| \gg 0$) are primarily processed by the power-law component $\lambda |z_i|^\alpha$ for suppression, whereas weak signals ($|z_i| \approx 0$) are enhanced through the modified tanh operation $(1 - \lambda) \frac{e^{\beta|z_i|} - 1}{e^{\beta|z_i|} + 1}$.

The SBB module employs a simple feature modulation function to progressively dilute channel-spatial dual-dimensional biases through layer-wise and point-wise operations. This feature modulation function demonstrates a smoothing effect on feature distributions by attenuating high-magnitude features while proportionally enhancing subtle low-magnitude characteristics.

Unlike traditional normalization or scale compression operations, the transformation function of SBB is specifically designed to address the "dual-bias" phenomenon revealed in Section 1: (1) dominant channels or spatial activation areas specific to the source domain often exhibit overconfident responses, but show significant performance degradation in the target domain; (2) the truly transferable general features are often distributed in the low-magnitude tail region, which can easily be overlooked by the model. To address this phenomenon, we design a parameter-free modulation function that uses a sublinear power-law function to compress high-magnitude activations, thereby suppressing the model's over-reliance on source-domain-specific responses; meanwhile, we use an enhanced tanh function to moderately amplify low-magnitude activations, uncovering potentially transferable low-magnitude features. To avoid distorting the original semantic space, we specifically design a sign-preserving mechanism, ensuring that the geometric structure of the features in the embedding space remains intact, thus reliably diluting source-domain bias without introducing any learnable parameters. The curve of our modulation function is shown in Fig. 3. It primarily reduces the relative differences between high- and low-magnitude values without altering the overall distribution trend. Consequently, the original discriminative features are preserved in a "weakened yet retained" manner. As shown in the magnitude distribution plot (Fig. 13) and the attention heatmap (Fig. 12), the original discriminative features are not significantly altered but appropriately attenuated, with their discriminative regions and characteristics remaining clearly observable. The overall distribution trend of the features remains unchanged. In other words, this transformation reduces the dynamic range of feature magnitudes while preserving the overall distribution trend, thereby achieving magnitude equalization without significantly impairing discriminative capability.

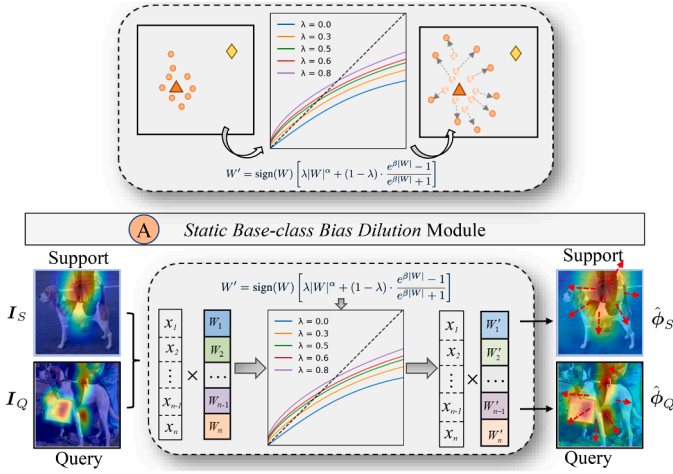


Fig. 3. Structure of the SBB module. The SBB employs feature modulation to dilute base-class biases, transforming input features (I_S , I_Q) into bias-suppressed representations ($\hat{\phi}_S$, $\hat{\phi}_Q$).

3.2. Dynamic novel-class bias recomposition module

Sections 3.2 and 3.3 present our dual-component bias recomposition framework: (1) the dynamic novel-class bias recomposition (DNBR) module generates target-adaptive channel-spatial dual soft masks via lightweight depthwise separable convolutions, and (2) the masked-gradient MAML (MG-MAML) strategy optimizes these masks through gradient-aware sparse parameter updates.

As shown in Fig. 1, our analysis reveals the dual inherent biases formed during source domain training: excessive reliance on source-specific feature responses in the channel dimension, and rigid preference for source-typical structures in the spatial dimension. Traditional methods often address channel-wise adaptation in isolation, overlooking the inherent coupling between these biases in cross-domain scenarios. Our key insight is that these two biases—channel-specific pattern dependency and spatial-specific structure preference—must be jointly calibrated through a unified framework. We reinterpret the kernels of depthwise separable convolutions as feature selection soft masks: depthwise convolutions act as spatial masks, and pointwise convolutions as channel masks, enabling dual-bias calibration. The framework of the DNBR module is illustrated in Fig. 4.

3.2.1. Depth-wise convolution as spatial mask

Spatial bias calibration requires accurate localization of domain-sensitive structural patterns. We reveal that depthwise convolution kernels intrinsically exhibit generalized spatial masking properties. By reparameterizing these kernels as learnable soft masks, we concurrently achieve precise spatial adaptation and parameter efficiency. We propose to reinterpret depthwise convolution as a spatial masking operation. The mathematical foundation is established as follows.

Mask Equivalence Proof. Given input features $\hat{\phi} \in \mathbb{R}^{C \times H \times W}$, standard 3×3 depthwise convolution performs:

$$\hat{\phi}_{\text{spatial}}^{(c,i,j)} = \sum_{m=-1}^1 \sum_{n=-1}^1 \theta_d^{(c,m,n)} \cdot \hat{\phi}^{(c,i+m,j+n)}, \quad (4)$$

where $\theta_d \in \mathbb{R}^{3 \times 3 \times 3}$ denotes depthwise convolution kernels. We formally prove this operation is equivalent to:

$$\hat{\phi}'_{\text{spatial}} = \mathbf{M}_s(\theta_d) \star \hat{\phi}, \quad (5)$$

here, $\star$ indicates broadcasted multiply-add operations, and $\mathbf{M}_s(\cdot)$ is an identity mapping from parameters to soft masks. The equivalence holds because each kernel $\theta_d^{(c)}$ uniquely corresponds to the spatial mask of channel c , and the sliding window operation applies position-dependent feature weighting, which ensures the alignment of the mask positions.

Fundamental Characteristics. The masking operation has three fundamental characteristics. First, it exhibits locality, as each 3×3 mask only affects local receptive fields. Second, it ensures channel isolation, meaning that the masks are applied independently to each channel. Lastly, the operation is parameter-efficient, requiring only $9C$ parameters to achieve full spatial modulation.

Gradient Propagation Analysis. The gradient computation confirms the mask interpretation:

$$\frac{\partial \mathcal{L}}{\partial \theta_d^{(c,m,n)}} = \sum_{i,j} \frac{\partial \mathcal{L}}{\partial \hat{\phi}_{\text{spatial}}^{(c,i,j)}} \cdot \hat{\phi}^{(c,i+m,j+n)}, \quad (6)$$

$$\frac{\partial \mathcal{L}}{\partial \hat{\phi}^{(c,i,j)}} = \sum_{m,n} \theta_d^{(c,m,n)} \cdot \frac{\partial \mathcal{L}}{\partial \hat{\phi}_{\text{spatial}}^{(c,i-m,j-n)}}. \quad (7)$$

This demonstrates that the gradient updates to the mask parameters are proportional to the intensity of the input feature activations. The updates are also localized to the corresponding spatial positions, ensuring that only relevant areas are adjusted. Additionally, the updates are channel-specific, with no interference between channels.

3.2.2. Point-wise convolution as channel mask

Channel-wise adaptation necessitates dynamic feature reweighting using inter-channel correlations. Unlike conventional channel attention mechanisms that process each channel independently, our method explicitly models inter-channel dependencies through the mask equivalence property of pointwise convolution (Fig. 4). This enables the model to suppress source-specific channels while enhancing transferable features in a context-aware manner.

Departing from the conventional view of pointwise convolution as mere feature transformation, we reinterpret it as a dynamic channel-wise masking mechanism. The implementation details are as follows.

Mask Equivalence Proof. Given spatially modulated features $\hat{\phi}'_{\text{spatial}} \in \mathbb{R}^{C \times H \times W}$, we process them with:

$$\hat{\phi}'_{\text{channel}} = \sum_{c=1}^C \theta_p^{(k,c)} \cdot \hat{\phi}'_{\text{spatial}}^{(c,i,j)}, \quad (8)$$

where $\theta_p \in \mathbb{R}^{C \times C \times 1 \times 1}$ denotes pointwise convolution kernels. We reinterpret this as channel masking:

$$\hat{\phi}'_{\text{channel}} = \mathbf{M}_c(\theta_p) \star \hat{\phi}'_{\text{spatial}}, \quad (9)$$

here, $\star$ indicates broadcasted multiply-add operations along channel dimension, and $\mathbf{M}_c(\cdot)$ is an identity mapping from parameters to masks. This equivalence reveals that each output channel k is associated with a set of input channel weights $\{\theta_p^{(k,c)}\}_{c=1}^C$. Furthermore, the weight values directly act as channel mask coefficients, without the need for any conversion.

Dynamic Mask Properties. Compared to static masks, this approach offers three advantages. First, it enables cross-channel interaction, allowing for nonlinear combinations through a $C \times C$ parameter matrix. Second, it maintains spatial consistency, as all points at the same spatial location share identical channel masking patterns. Finally, it supports end-to-end learning, with mask coefficients being automatically optimized through backpropagation.

Gradient Propagation Analysis. The gradient computation now reflects the cascaded operations:

$$\frac{\partial \mathcal{L}}{\partial \theta_p^{(k,c)}} = \sum_{i,j} \frac{\partial \mathcal{L}}{\partial \hat{\phi}'_{\text{channel}}^{(k,i,j)}} \cdot \hat{\phi}'_{\text{spatial}}^{(c,i,j)}, \quad (10)$$

$$\frac{\partial \mathcal{L}}{\partial \hat{\phi}'_{\text{spatial}}^{(c,i,j)}} = \sum_{k=1}^C \theta_p^{(k,c)} \cdot \frac{\partial \mathcal{L}}{\partial \hat{\phi}'_{\text{channel}}^{(k,i,j)}}. \quad (11)$$

This shows mask coefficient updates are now conditioned on spatially modulated features, creating a hierarchical adaptation mechanism.

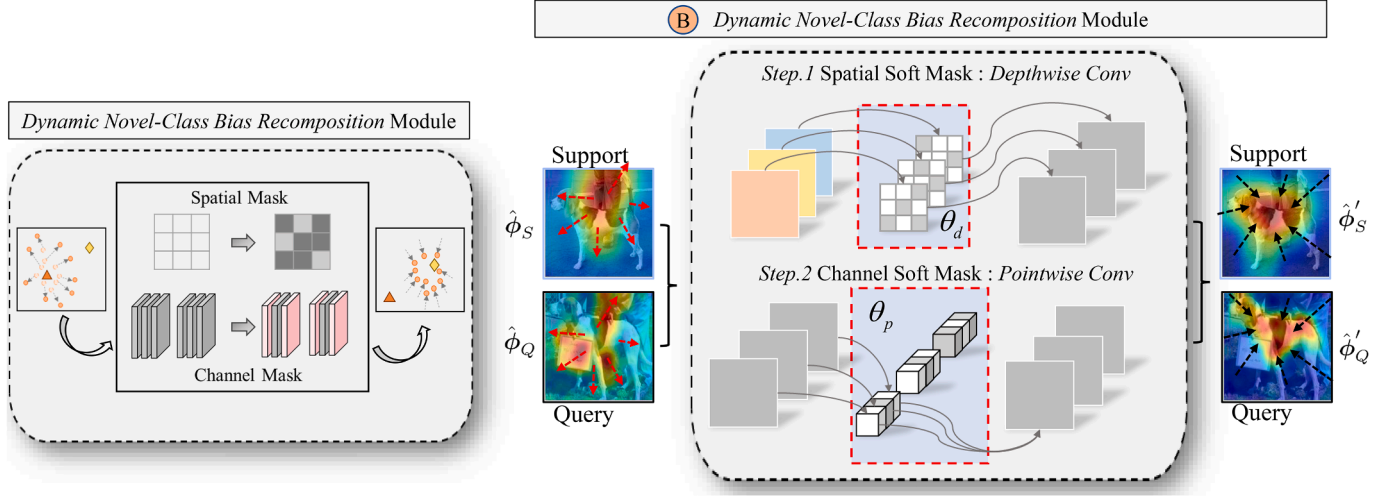


Fig. 4. Structure of the DNBR module. The DNBR employs meta-optimized lightweight depthwise separable convolutions to generate target-adaptive channel importance weights and spatial preference weights, enabling coordinated optimization of dual biases. The left panel illustrates the overall architecture. The right panel details the two-step implementation: (1) depthwise convolution generates spatial soft masks to adjust region preferences, followed by (2) pointwise convolution producing channel soft masks for channel reweighting.

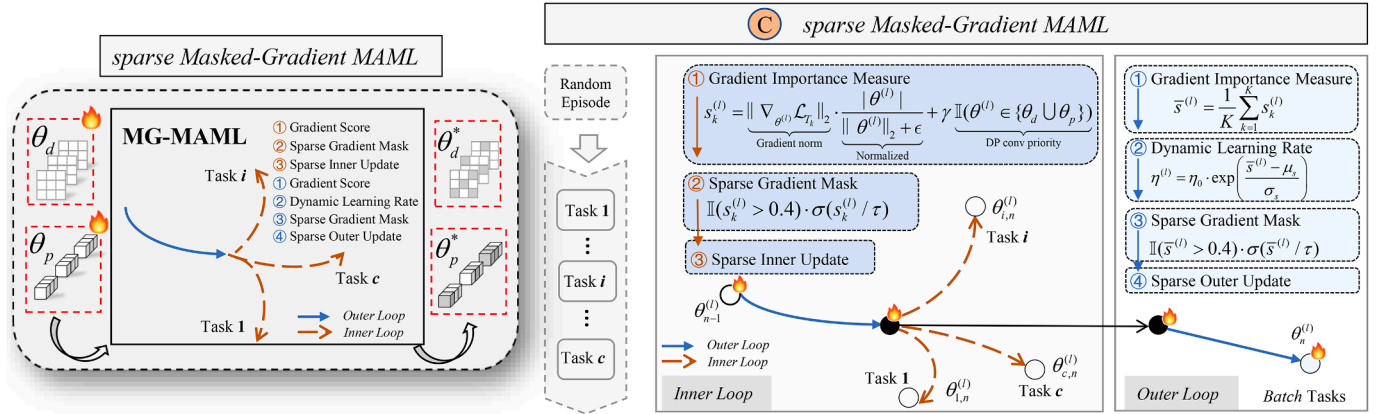


Fig. 5. Detailed schematic of MG-MAML's optimization. Left: Overall architecture showing the dual-loop optimization process, where inner loop performs task-specific sparse updates (blue arrows) and outer loop conducts cross-task meta-updates (orange arrows). Right: Detailed workflow of gradient sparsification: (1) gradient importance measurement combines gradient norms and parameter magnitudes; (2) dynamic sparse masks selectively update depthwise (θ_d) and pointwise (θ_p) convolution kernels; (3) task-adaptive learning rates scale with parameter importance.

3.2.3. Residual learning and training stability

To preserve the original feature representation, we employ gated residual connections that directly link to the initial input $\hat{\phi}$:

$$\hat{\phi}'_{\text{out}} = \sigma(v) \cdot \hat{\phi}'_{\text{channel}} + (1 - \sigma(v)) \cdot \hat{\phi}, \quad (12)$$

where v is initialized to -1.0 (corresponding to $\sigma(v) = 0.27$) and learned via:

$$\frac{\partial \mathcal{L}}{\partial v} = \sum_{c,i,j} \frac{\partial \mathcal{L}}{\partial \hat{\phi}^{(c,i,j)}} \cdot \sigma'(v) \cdot (\hat{\phi}^{(c,i,j)}_{\text{channel}} - \hat{\phi}^{(c,i,j)}). \quad (13)$$

3.3. Soft mask tuning via masked-gradient MAML

Leveraging meta-tasks built via episodic sampling, we propose masked-gradient MAML (MG-MAML), a novel framework that integrates gradient magnitude-aware parameter selection with the meta-optimization of depthwise separable convolutions. This approach addresses two critical limitations of conventional MAML: (1) the global parameter update paradigm that ignores task-specific characteristics, and (2) the risk of channel-spatial dual-mask overfitting caused by uniform gradient propagation. Through a dynamic sparsification mecha-

nism that selectively updates key spatial masks (depthwise kernels) and channel masks (pointwise kernels), the framework adaptively optimizes both channel reweighting and spatial structure adjustment processes in depthwise separable convolutions.

Methodological Innovations. Fig. 5 demonstrates three key advancements beyond conventional MAML [45]: (1) Gradient Magnitude-aware Sparsity dynamically identifies critical parameters using task-specific importance metrics combining gradient and parameter magnitudes Eq. (14). (2) Sparse Depthwise Separable Convolution Update: apply a sparse update strategy to the dual-mask parameters of depthwise separable convolutions—spatial masks (depthwise kernels, θ_d) and channel masks (pointwise kernels, θ_p). (3) Task-adaptive Meta Learning automatically modulates meta-update strengths through cross-task importance accumulation Eq. (16).

Dynamic Parameter Selection. Our importance-driven update mechanism evaluates each parameter $\theta^{(l)}$ in task T_k through a composite scoring function:

$$s_k^{(l)} = \underbrace{\|\nabla_{\theta^{(l)}} \mathcal{L}_{T_k}\|_2}_{\text{Gradient norm}} \cdot \underbrace{\frac{|\theta^{(l)}|}{\|\theta^{(l)}\|_2 + \epsilon}}_{\text{Normalized magnitude}} + \underbrace{\gamma \mathbb{I}(\theta^{(l)} \in \{\theta_d \cup \theta_p\})}_{\text{DP conv priority}}, \quad (14)$$

with $\epsilon = 10^{-5}$ ensuring numerical stability and $\gamma = 0.2$ controlling the prioritization toward depthwise separable convolution parameters. The task loss $\mathcal{L}_{T_k}$ drives gradient computation, while the indicator function $\mathbb{I}(\cdot)$ ensures that depthwise (θ_d) and pointwise (θ_p) convolution kernels receive higher update probabilities during meta-adaptation.

Sparse Mask-Guided Inner Loop. Task-specific adaptation employs a gated update mechanism that generates sparse masks based on importance scores:

$$\theta_{k,n}^{(l)} = \theta_{k,n-1}^{(l)} - \eta \cdot \underbrace{\mathbb{I}(s_k^{(l)} > 0.4) \cdot \sigma(s_k^{(l)} / \tau)}_{\text{Sparse mask}} \odot \nabla_{\theta_{k,n-1}^{(l)}} \mathcal{L}_{T_k}, \quad (15)$$

where $\theta_{k,n}^{(l)}$ represents the parameter state after n updates in meta-task k . The element-wise product $\odot$ combines gradients with sigmoid-weighted masks ($\tau = 0.15$) that maintain gradient flow for moderately important parameters while freezing updates for low-scoring components ($s_k^{(l)} \leq 0.4$). The sparse mask in Eq. (15) implements hierarchical sparsity: depthwise and pointwise kernels with scores below 0.4 are frozen to preserve source-domain knowledge, while active kernels undergo sigmoid-weighted updates. This ensures that only the most transfer-relevant parameters in both spatial and channel dimensions are adapted, maintaining the decoupled advantage of depthwise separable convolutions.

Outer Loop Meta-Update. The meta-optimization phase enhances gradient updates through two key modifications. First, it aggregates parameter importance across tasks using $\bar{s}^{(l)}$, computed as the mean importance score for $\theta^{(l)}$ across K meta-tasks. Second, it implements an adaptive learning rate scheme where $\eta^{(l)}$ scales proportionally with each parameter's cross-task significance, prioritizing updates for consistently relevant parameters. These components are formally expressed as:

$$\bar{s}^{(l)} = \frac{1}{K} \sum_{k=1}^K s_k^{(l)}, \quad \eta^{(l)} = \eta_0 \cdot \exp\left(\frac{\bar{s}^{(l)} - \mu_s}{\sigma_s}\right), \quad (16)$$

where μ_s and σ_s denote the global mean and standard deviation of importance scores. The exponential formulation achieves three objectives: (1) amplifying updates for parameters with stable importance across tasks, (2) suppressing gradient contributions from inconsistent dimensions, and (3) automatically adapting to diverse task characteristics. This sparse meta-update mechanism operates in conjunction with the previously described masking strategy, with complete implementation specifics provided in Algorithm 1. The outer-loop updates also utilize a sparse strategy. For details, please refer to Algorithm 1.

3.4. Novel-class cross-image semantic alignment module

Following novel-class bias recomposition, Our novel-class cross-image semantic alignment (NCSA) module establishes dual-dimensional correspondences between query-support pairs, where channel-wise correlation and spatial correspondence work in concert to: (1) jointly en-

Algorithm 1: MG-MAML for parameter tuning.

Require: Meta tasks $\{T_k\}_{k=1}^K$, initial parameters θ_0

Require: Base learning rate η_0 , $\tau = 0.15$, $\kappa = 0.4$

```

1 for each meta-iteration do
2   for each task  $T_k \sim \{T_k\}_{k=1}^K$  do
3     Inner Loop Adaptation:
4     for each parameter  $\theta^{(l)} \in \theta$  do
5       Compute importance score (Eq. 14):
6
7        $s_k^{(l)} = \|\nabla_{\theta^{(l)}} \mathcal{L}_{T_k}\|_2 \cdot \frac{|\theta^{(l)}|}{\|\theta^{(l)}\|_2 + \epsilon} + \gamma \mathbb{I}(\theta^{(l)} \in \{\theta_d \cup \theta_p\})$ 
8       Generate sparse mask:
9        $M_k^{(l)} = \mathbb{I}(s_k^{(l)} > \kappa) \cdot \sigma(s_k^{(l)} / \tau)$ 
10      Update parameters:
11       $\theta_k^{(l)} \leftarrow \theta_0^{(l)} - \eta_0 \cdot M_k^{(l)} \odot \nabla_{\theta_0^{(l)}} \mathcal{L}_{T_k}$ 
12    end for
13  Outer Loop Meta-Update:
14  Compute cross-task score and LR (Eq. 16):
15   $\bar{s}^{(l)} = \frac{1}{K} \sum_{k=1}^K s_k^{(l)}$ 
16   $\eta^{(l)} = \eta_0 \cdot \exp\left(\frac{\bar{s}^{(l)} - \mu_s}{\sigma_s}\right)$ 
17  for each parameter  $\theta^{(l)} \in \theta$  do
18    Meta-update with sparse mask:
19     $\theta_0^{(l)} \leftarrow \theta_0^{(l)} - \eta^{(l)} \cdot \mathbb{I}(\bar{s}^{(l)} >$ 
20     $\kappa) \cdot \sigma(\bar{s}^{(l)} / \tau) \odot \left(\frac{1}{K} \sum_{k=1}^K \nabla_{\theta_0^{(l)}} \mathcal{L}_{T_k}^{\text{val}}\right)$ 
21  end for
22 Output: Optimized parameters  $\theta^*$  and soft mask parameters  $\theta_p^*$  and  $\theta_d^*$ .

```

hance feature discriminability through cross-dimensional pattern reinforcement, and (2) cooperatively ensure semantic consistency via unified channel-spatial alignment. This cross-image semantic alignment strategy effectively transfers target-specific patterns while suppressing domain-specific noise, enabling superior few-shot adaptation under domain shifts.

The schematic diagram of NCSA is illustrated in Fig. 6. Given input images—support image I_S and query image I_Q —the processed feature maps after {SBB, DNBR, NCSA} are denoted as $\{\hat{\phi}_S'$ and $\hat{\phi}_Q'$, $\hat{\phi}_S''$ and $\hat{\phi}_Q''\}$.

Fig. 6 demonstrates the initial cross-image correlation computation between $\hat{\phi}_S'$ and $\hat{\phi}_Q'$, which guides subsequent feature alignment. Both

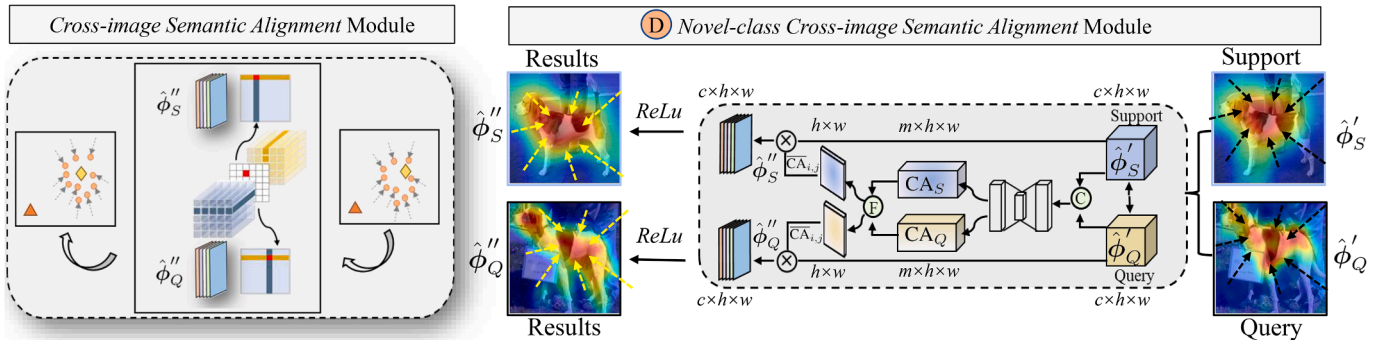


Fig. 6. Structure of the NCSA module. The bias-recomposed features $\hat{\phi}_S'$ and $\hat{\phi}_Q'$ undergo support-query cross-image alignment to obtain the aligned features $\hat{\phi}_S''$ and $\hat{\phi}_Q''$, significantly improving target-domain feature discriminability and semantic consistency.

feature maps are first reshaped into $\mathbb{R}^{C \times m}$ matrices:

$$\hat{\phi}'_S = [\tilde{\phi}'_S, \tilde{\phi}'_S, \dots, \tilde{\phi}'_S], \quad \hat{\phi}'_Q = [\tilde{\phi}'_Q, \tilde{\phi}'_Q, \dots, \tilde{\phi}'_Q], \quad (17)$$

Where $m = H \times W$ represents the total spatial positions in the feature maps, with W and H denoting width and height dimensions respectively, and C indicating the channel depth. The feature vectors $\tilde{\phi}'_S$ and $\tilde{\phi}'_Q$, corresponding to spatial location i in $\hat{\phi}'_S$ and $\hat{\phi}'_Q$ respectively, are analyzed for their semantic relationships. We establish their semantic correspondence by computing pairwise cosine similarities between all vector combinations, resulting in a comprehensive correlation matrix that systematically captures cross-position feature relationships across the entire spatial extent of both representations. The correlation matrix is computed as follows:

$$CA_{i,j} = \frac{\langle \tilde{\phi}'_Q, \tilde{\phi}'_S \rangle}{\|\tilde{\phi}'_Q\| \|\tilde{\phi}'_S\|}, \quad i, j = 1, \dots, m, \quad (18)$$

where $\langle \cdot, \cdot \rangle$ denotes inner product and $\|\cdot\|$ represents Euclidean norm. The correlation matrix $CA_{i,j}$ captures channel-spatial relationships between support and query features.

Fusion Embedding Generation. This stage generates cross-aligned features by spatially reweighting according to $CA_{i,j}$. Support feature adaptation employs row-wise softmax normalization of the correlation matrix (query feature adaptation follows a symmetric process):

$$\overline{CA}_{i,j} = \frac{\exp(CA_{i,j})}{\sum_{i=1}^{HW} \exp(CA_{i,j})}. \quad (19)$$

The resulting aligned support feature $\hat{\phi}''_S \in \mathbb{R}^{C \times m}$ is obtained through matrix multiplication between $\overline{CA}$ and the transposed support feature (with $\hat{\phi}''_Q$ computed analogously):

$$\hat{\phi}''_S = (\overline{CA} \cdot \hat{\phi}'_S)^T. \quad (20)$$

Algorithm 2 details the complete CSDBC workflow. Our framework systematically calibrates source-domain bias through progressive dilution, recombination, and alignment. First, the SBB module suppresses source-domain biases via parameter-free feature smoothing. Then, the DNBR module bridges domain gaps by generating target-adaptive channel-spatial soft masks using lightweight depthwise separable convolutions. Finally, the NCSA module calibrates target-domain features through dual-dimensional cross-image alignment. Together, these components form a comprehensive computational core Eqs. (1), (4), (8), (12), (15) and (18)–(20) that enables dynamic feature adaptation without explicit domain shift modeling. This hierarchical approach - sequentially performing bias suppression, adaptive recombination, and cross-image alignment - allows our system to automatically emphasize task-relevant features across varying domains while maintaining robust generalization capability, even with imperfect domain shift estimation.

4. Experimental result and analysis

This section comprehensively evaluates the CSDBC framework through: (1) ablation studies of individual components, (2) performance stability assessment under domain shifts, and (3) visualization of critical findings.

4.1. Implementation details

Dataset Overview. We follow standard protocols for CD-FSL evaluation, using *mini*-ImageNet [46] as the source domain with 64 base classes for training. The generalization capability is rigorously tested across eight distinct target domains that exhibit varying degrees of distributional shift. These include four natural image benchmarks (CUB [47],

Algorithm 2: Algorithmic description of CSDBC.

Require: Support image I_S , query image I_Q

Require: Mixing parameter λ

```

1 Stage 1: Static base-class bias dilution
2 for each image pair  $\{I_S, I_Q\}$  do
3   Obtain deep features  $\mathbf{z} = [z_1, \dots, z_m]$ 
4   for each component  $z_i \in \mathbf{z}$  do
5     Implement transformation via:
6      $\tilde{\phi}_\lambda(z_i) = \text{sign}(z_i) \left[ \lambda |z_i|^{0.5} + (1 - \lambda) \frac{e^{1.5|z_i|-1}}{e^{1.5|z_i|+1}} \right]$ 
7   end for
8   Construct diluted representation:
9    $\hat{\phi}_\lambda(\mathbf{z}) = [\tilde{\phi}_\lambda(z_1), \dots, \tilde{\phi}_\lambda(z_m)]$ 
10 end for
11 Stage 2: Dynamic novel-class bias recombination
12 Initialize depthwise kernels  $\theta_d$ , pointwise kernels  $\theta_p$ 
13 for each feature map  $\hat{\phi} \in \{\hat{\phi}_S, \hat{\phi}_Q\}$  do
14   Spatial soft masking:
15    $\hat{\phi}'_{\text{spatial}} \leftarrow \mathbf{M}_s(\theta_d) \star \hat{\phi}$ 
16   Channel soft masking:
17    $\hat{\phi}'_{\text{channel}} \leftarrow \mathbf{M}_c(\theta_p) \star \hat{\phi}'_{\text{spatial}}$ 
18   Generate recomposed features:
19    $\hat{\phi}' \leftarrow \sigma(v) \hat{\phi}'_{\text{channel}} + (1 - \sigma(v)) \hat{\phi}$ 
20 end for
21 Stage 3: Novel-class cross alignment
22 for feature maps  $\{\hat{\phi}'_S, \hat{\phi}'_Q\}$  do
23   Reshape  $\hat{\phi}'_S, \hat{\phi}'_Q$  to  $\mathbb{R}^{C \times m}$ 
24   Establish cross-image affinities:
25    $CA_{i,j} = \frac{\tilde{\phi}'_Q \cdot \tilde{\phi}'_S}{\|\tilde{\phi}'_Q\|_2 \|\tilde{\phi}'_S\|_2}, \quad \forall i, j \in \{1, \dots, m\}$ 
26   Normalize attention weights:
27    $\overline{CA}_{i,j} \leftarrow \text{softmax}(CA_{i,j})$ 
28   Generate aligned features:
29    $\hat{\phi}'' = (\overline{CA} \cdot \hat{\phi}'^T)^T$ 
30 end for
31 Output: Aligned features  $\{\hat{\phi}''_S, \hat{\phi}''_Q\}$ .
```

Table 1

Cross-domain evaluation datasets.

Domain shift	Dataset	Description	Classes
Natural near-domain	CUB [47]	Fine-grained birds	50
	Cars [48]	Fine-grained cars	49
	Plantae [50]	Plantae images	50
	Places [49]	Scene images	19
Extreme distant-domain	CropDiseases [54]	Crop disease images	38
	EuroSAT [53]	Satellite images	10
	ISIC [52]	Skin lesion images	7
	ChestX [51]	X-ray images	7

Cars [48], Places [49], Plantae [50]) representing moderate domain gaps and four specialized visual domains (ChestX [51], ISIC [52], EuroSAT [53], CropDisease [54]) with substantially different data characteristics. As Table 1 shows, these benchmarks cover diverse tasks including fine-grained classification, medical diagnosis, and remote sensing analysis. Model selection is performed exclusively on the *mini*-ImageNet validation split prior to cross-domain assessment, ensuring no target-domain information leakage during training or hyperparameter tuning.

To gain a deeper understanding of the distributional divergence between the source and target domains, we conduct a quantitative anal-

Table 2

Quantitative analysis of distribution divergence between *near*-domain and *distant*-domain datasets. Here, *mini* represents *mini*-ImageNet, KLD represents KL Divergence, and WD represents Wasserstein Distance.

Dataset Pair	MMD	KLD	WD
<i>mini</i> vs CUB [47]	0.15	0.20	0.32
<i>mini</i> vs Cars [48]	0.18	0.24	0.45
<i>mini</i> vs Places [49]	0.14	0.17	0.30
<i>mini</i> vs Plantae [50]	0.17	0.22	0.43
<i>mini</i> vs ChestX [51]	0.25	0.32	0.57
<i>mini</i> vs ISIC [52]	0.30	0.35	0.62
<i>mini</i> vs EuroSAT [53]	0.18	0.26	0.48
<i>mini</i> vs CropDisease [54]	0.20	0.28	0.52
Avg (First 4)	0.16	0.21	0.38
Avg (Last 4)	0.23	0.30	0.55

ysis of the distributional differences between the source domain (*mini*-ImageNet) and eight target domain datasets. Specifically, we use three metrics-MMD [27], KL Divergence [55], and Wasserstein Distance [56]-to assess the distributional differences between *near*-domain and *distant*-domain datasets. The degree of domain shift is presented in Table 2, showing significant variation in the distributional divergence between the source domain and different target domains. Specifically, datasets commonly considered as “*natural* domains”, such as CUB [47], Cars [48], Places [49], and Plantae [50], exhibit smaller distributional differences with the source domain; whereas datasets regarded as “*extreme* domains”, such as ChestX [51], ISIC [52], EuroSAT [53], and CropDisease [54], demonstrate considerably larger distributional differences. Based on these findings, we classify CUB, Cars, Places, and Plantae as *near*-domain datasets, and ChestX, ISIC, EuroSAT, and CropDisease as *distant*-domain datasets.

Benchmark Comparisons. Our CSDBC framework is rigorously compared against three distinct methodological paradigms to demonstrate its cross-domain effectiveness. First, we examine conventional FSL techniques [8,46] utilizing episodic training strategies. Second, we contrast with contemporary CD-FSL approaches [6,19,22,23,54,57] specifically designed for domain adaptation scenarios. Third, following insights from Guo et al. [5] about the superior performance of fine-tuning methods under domain shifts, we include competitive pretraining-finetuning baselines [5,19,20,57]. This tripartite evaluation strategy thoroughly assesses our novel bias calibration framework - particularly the base-class dilution and novel-class recomposition components - across varying levels of domain divergence.

4.2. Implementation details

Training Configuration. We adopt ResNet-10 [5,19,57] as our base backbone, initialized with weights pretrained on the *mini*-ImageNet source dataset. Consistent with prior work [23], input images are standardized to 224×224 . Optimization is performed using Adam with an initial learning rate of 0.001 and momentum parameter of 0.9. We conduct complete end-to-end training utilizing the PyTorch deep learning framework, with all experiments running on NVIDIA 2080 Ti GPU computing platforms.

Evaluation Protocol. Following the standard CD-FSL evaluation framework [23,57], we use the conventional n -way k -shot paradigm with 15 query samples per class. To ensure statistical reliability, we evaluate model performance across 600 randomly generated test episodes per target domain, examining both 1-shot and 5-shot learning configurations. The $top-1$ classification accuracy serves as our primary metric, providing a comprehensive assessment of the model’s ability to generalize across domains while facilitating direct comparison with state-of-the-art approaches under identical experimental conditions.

4.3. Experiments on dual-bias validation

4.3.1. Spatial bias: Region definition and sensitivity analysis

The Definition and Size of Region. In our study, regions are sampled from the downsampled feature maps. Specifically, we compute cross-domain distributional differences at a 1×1 region granularity, i.e., at the level of a single spatial unit. Each element $[C, i, j]$ in the feature map represents an abstract representation of a receptive field region in the input image. For a feature map of size $[C \times H \times W]$, we partition it into $H \times W$ non-overlapping sub-regions (regions), each corresponding to a 1×1 spatial window. The cross-domain discrepancy (S-MMD) is then computed based on the feature distribution of each region.

Comparative Analysis of Different Region Sizes. As shown in Table 3, we design several region partition strategies in the spatial bias analysis to examine the influence of region granularity on cross-domain performance discrepancies. On the high-dimensional feature maps, we adopt four overlapping partition schemes: pixel-wise partitioning (1×1), small-region partitioning (2×2 , 3×3), and large-region partitioning (5×5). The experimental results show that source-specific regions consistently exhibit significant cross-domain performance degradation under all granularities, with the gap between the source and target domains always exceeding 10%. For example, under the 1×1 granularity, the accuracy of source-specific regions drops from 56.01% in the source domain to 41.68% in the target domain, yielding a gap of 14.33%. At the 3×3 granularity, the accuracy decreases from 53.10% to 41.85%, with a gap of 11.25%. Even at the 5×5 granularity, the performance gap of source-specific regions still reaches 10.81%. In contrast, non-source-specific regions maintain relatively stable performance across different partition scales, with the gap usually smaller than 3% (e.g., 2.11% under 1×1 , further reduced to 0.86% under 5×5). These results indicate that cross-domain degradation mainly occurs in source-specific regions, whereas non-source-specific regions are more robust to domain transfer. Moreover, the degradation is not sensitive to the choice of partition granularity, suggesting that the bias does not originate from the partition strategy itself but rather from the inherent source-specific spatial bias in the feature representations. In particular, smaller granularities (e.g., 1×1) reveal more fine-grained spatial discrepancies, thereby amplifying performance drops in source-specific regions; larger granularities (e.g., 5×5), while smoothing out details to some extent, still clearly highlight the significant bias in source-specific regions.

4.3.2. Generality validation of the dual-bias theory

To further validate the generality of the dual-bias theory, we conduct extensive experiments across different backbones and source domain datasets. Through these experiments, we explore the stability and universality of the dual-bias phenomenon under various network architectures and source domain settings. The experimental results show that dual-bias is not limited to the ResNet-10 [5,19,57] used in the original study, and it stably exists across other backbones and different source domain datasets, further demonstrating the generality of the dual-bias theory in CD-FSL tasks. In our experiments, channels are divided based on their C-MMD values into non-source-specific channels (the 20% of channels with the lowest C-MMD values) and source-specific channels (the remaining channels); similarly, regions are divided based on their S-MMD values into non-source-specific regions (the 20% of regions with the lowest S-MMD values) and source-specific regions (the remaining regions).

Cross-backbone Generality Validation. We conduct experiments on multiple mainstream few-shot learning networks, including Conv-4, ResNet-10, and ResNet-12. The experimental results show that both channel-bias and spatial-bias stably exist, whether at the channel level (see Table 4) or at the spatial level (see Table 5), proving that the dual-bias phenomenon is not limited to ResNet-10 used in the original study. Specifically, under these backbones, source-specific features exhibit higher accuracy in the source domain but suffer from signifi-

Table 3

Performance comparison on source-specific and non-source-specific regions under different region sizes. All values represent percentages (%). Non-source-specific regions are defined as the 20% of regions with the lowest S-MMD values, while source-specific regions are defined as the remaining regions.

Region Size	Partition Strategy	Region Type	Src Acc	Tgt Acc	Gap ↓
1 × 1	Pixel-wise	Source-specific	56.01	41.68	14.33
		Non-source-specific	50.07	47.96	2.11
2 × 2	Sliding window (stride 1, overlapping)	Source-specific	53.34	40.90	12.44
		Non-source-specific	47.96	45.90	2.06
3 × 3	Sliding window (stride 1, overlapping)	Source-specific	53.10	41.85	11.25
		Non-source-specific	47.72	46.23	1.49
5 × 5	Sliding window (stride 1, overlapping)	Source-specific	53.73	42.92	10.81
		Non-source-specific	48.03	47.17	0.86

Table 4

Generality validation of dual-bias characterization across different backbones and source domains (channel-level). All values represent percentages (%). Non-source-specific channels are defined as the 20% of channels with the lowest C-MMD values, while source-specific channels are defined as the remaining channels. In all datasets, source-specific channels achieve higher source-domain accuracy but suffer severe degradation in the target domain, while non-source-specific channels maintain more stable performance.

(a) <i>tiered</i> -ImageNet as Source Domain			
Backbone	Channel Type	Src Acc	Tgt Acc
Conv-4	Source-specific	44.25	33.30
	Non-source-specific	41.40	37.70
ResNet-10	Source-specific	55.62	44.25
	Non-source-specific	52.10	48.35
ResNet-12	Source-specific	58.05	46.10
	Non-source-specific	54.25	50.50
(b) iNaturalist as Source Domain			
Backbone	Channel Type	Src Acc	Tgt Acc
Conv-4	Source-specific	41.70	31.55
	Non-source-specific	38.90	35.25
ResNet-10	Source-specific	50.08	40.00
	Non-source-specific	47.65	44.20
ResNet-12	Source-specific	51.50	39.85
	Non-source-specific	47.75	44.05
(c) EuroSAT as Source Domain			
Backbone	Channel Type	Src Acc	Tgt Acc
Conv-4	Source-specific	40.10	29.15
	Non-source-specific	37.35	33.95
ResNet-10	Source-specific	49.95	38.70
	Non-source-specific	46.85	43.40
ResNet-12	Source-specific	50.25	38.25
	Non-source-specific	48.60	42.80

Table 5

Generality validation of dual-bias characterization across different backbones and source domains (spatial-level). All values represent percentages (%). Non-source-specific regions are defined as the 20% of regions with the lowest S-MMD values, while source-specific regions are defined as the remaining regions. In all datasets, source-specific regions achieve higher source-domain accuracy but suffer severe degradation in the target domain, while non-source-specific regions maintain more stable performance.

(a) <i>tiered</i> -ImageNet as Source Domain			
Backbone	Region Type	Src Acc	Tgt Acc
Conv-4	Source-specific	43.10	31.80
	Non-source-specific	39.95	36.20
ResNet-10	Source-specific	56.20	43.10
	Non-source-specific	52.85	48.30
ResNet-12	Source-specific	57.35	45.25
	Non-source-specific	54.50	50.60
(b) biNaturalist as Source Domain			
Backbone	Region Type	Src Acc	Tgt Acc
Conv-4	Source-specific	40.60	29.05
	Non-source-specific	37.20	33.50
ResNet-10	Source-specific	49.75	37.20
	Non-source-specific	46.05	42.40
ResNet-12	Source-specific	51.95	39.00
	Non-source-specific	48.00	44.10
(c) EuroSAT as Source Domain			
Backbone	Region Type	Src Acc	Tgt Acc
Conv-4	Source-specific	39.65	28.40
	Non-source-specific	36.50	33.10
ResNet-10	Source-specific	49.25	37.35
	Non-source-specific	46.10	42.55
ResNet-12	Source-specific	50.40	38.00
	Non-source-specific	46.70	42.85

cant performance degradation in the target domain, while non-source-specific features maintain more stable performance across domains. These results validate the generality of dual-bias across different backbones.

Cross-source Domain Generality Validation. In addition to *mini*-ImageNet, we also validate on several representative source domain datasets, including real-object category dataset *tiered*-ImageNet [58], natural-object category dataset iNaturalist [50], and remote-sensing object category dataset EuroSAT [53] (see Tables 4 and 5). On these different source domain datasets, we also observe consistent dual-bias patterns. Source-specific features typically achieve higher accuracy in the source domain but show a significant drop in performance on the target domain, while non-source-specific features are relatively more stable. This further proves that dual-bias characterization is not dependent on

specific source domain settings and is universally applicable across different source domain datasets.

4.4. Ablation studies

For our ablation analysis, we utilize ResNet-10 [5,19,57] as the base architecture within the 5-way n -shot setting. The systematic investigation presented in Table 6 examines the contributions of four key modules (SBBD, DNBR, MG-MAML, NCSA) both individually and in various combinations. This evaluation spans eight distinct cross-domain datasets, which we further classify according to their domain shift characteristics into three groups: *near*-domain for minimal shifts, *distant*-domain for substantial shifts, and *full*-domain encompassing all test scenarios, enabling thorough assessment across different levels of domain shift.

Table 6

Ablation study on 5-way 5-shot tasks using *mini*-ImageNet as source. Best performance is highlighted in **bold** (mean $\pm$ S.D.%). All values represent percentages (%). The symbol ($\checkmark$) denotes active components.

Baseline	SBB	DNBR* (original MAML)	DNBR (MG-MAML)	NCSA	Ave: <i>near</i> -domain	Ave: <i>distant</i> -domain	Average
$\checkmark$					58.50 $\pm$ 0.31	57.10 $\pm$ 0.27	57.80 $\pm$ 0.29
$\checkmark$	$\checkmark$				61.40 $\pm$ 0.48	61.12 $\pm$ 0.21	61.26 $\pm$ 0.35
$\checkmark$		$\checkmark$			61.50 $\pm$ 0.56	61.27 $\pm$ 0.52	61.39 $\pm$ 0.54
$\checkmark$			$\checkmark$		62.46 $\pm$ 0.25	62.18 $\pm$ 0.32	62.32 $\pm$ 0.29
$\checkmark$				$\checkmark$	61.90 $\pm$ 0.57	61.78 $\pm$ 0.49	61.84 $\pm$ 0.54
$\checkmark$	$\checkmark$	$\checkmark$			62.40 $\pm$ 0.59	61.79 $\pm$ 0.48	62.10 $\pm$ 0.52
$\checkmark$	$\checkmark$		$\checkmark$		64.02 $\pm$ 0.23	63.79 $\pm$ 0.28	64.00 $\pm$ 0.26
$\checkmark$	$\checkmark$			$\checkmark$	62.66 $\pm$ 0.54	62.53 $\pm$ 0.51	62.60 $\pm$ 0.53
$\checkmark$		$\checkmark$		$\checkmark$	63.18 $\pm$ 0.56	63.01 $\pm$ 0.55	63.10 $\pm$ 0.56
$\checkmark$			$\checkmark$	$\checkmark$	64.49 $\pm$ 0.22	63.94 $\pm$ 0.29	64.22 $\pm$ 0.26
$\checkmark$	$\checkmark$	$\checkmark$		$\checkmark$	64.10 $\pm$ 0.61	63.59 $\pm$ 0.57	63.85 $\pm$ 0.59
$\checkmark$	$\checkmark$		$\checkmark$	$\checkmark$	64.86 $\pm$ 0.43	64.67 $\pm$ 0.45	64.77 $\pm$ 0.44

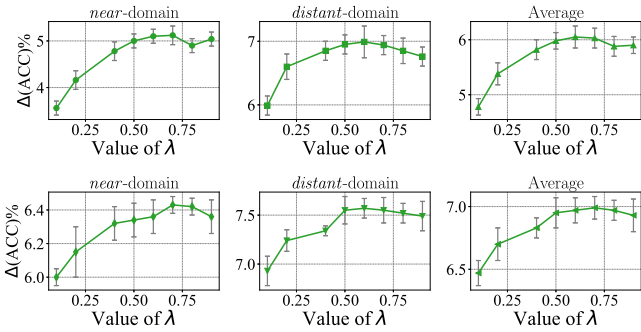


Fig. 7. Hyperparameter analysis of λ -values. Classification performance is evaluated across three domain configurations (*near*, *distant*, and *full*) with varying λ . The peak accuracy occurs at different λ values depending on domain shift magnitude, typically ranging from 0.5 to 0.7. Results are presented separately for 5-way 1-shot (top) and 5-way 5-shot (bottom) learning scenarios.

In Table 6, a checkmark “ $\checkmark$ ” indicates that a specific module is included in the corresponding cell, while an empty cell indicates its absence. Multiple checkmarks “ $\checkmark$ ” appearing together represent the simultaneous use of several modules. The notation DNBR* refers to the incomplete DNBR module implemented with MAML, whereas DNBR denotes the complete version employing our proposed MG-MAML.

The Influence of SBB. As shown in Table 6, removing the SBB module leads to significant performance degradation, particularly in distant domain shifts. This confirms SBB’s critical role in mitigating source-domain overfitting. Without SBB, models maintain strong preferences for source-specific features (e.g., natural image textures) but fail to adapt to target-domain structures (e.g., contours in satellite imagery). The impact weakens in *near*-domain transfers due to smaller bias gaps. Through parameter-free feature modulation, SBB effectively dilutes channel-wise and spatial source biases without computational overhead. Heatmap visualizations (Fig. 13) demonstrate SBB’s success in suppressing erroneous attention.

The Influence of The λ Value. The analysis of hyperparameter λ in Eq. (1) reveals its critical role in CD-FSL. As shown in Fig. 7, λ in the SBB module exhibits a clear “peak-then-decline” impact, with optimal values between 0.5–0.7 (final $\lambda=0.6$). This demonstrates λ ’s domain-dependent nature: smaller values (0.5) enhance feature smoothing for *distant*-domain tasks to mitigate source bias, while larger values (0.7) preserve transferable source knowledge in *near*-domain transfers. These findings align with Eq. (1)’s mathematical formulation, where λ balances source knowledge retention and target adaptation through tunable bias suppression.

The Influence of DNBR. The ablation study of the complete DNBR module (Table 6) demonstrates its superior performance across all benchmarks, particularly in *distant*-domain tasks (+5.08%). These re-

Table 7

One-Factor ablation study on the effect of hyperparameters κ , γ , and τ (5-way 5-shot).

(a) Effect of κ	
κ	Acc (%)
0.1	63.14
0.2	63.31
0.4	64.77
0.6	64.42
0.8	60.15
0.9	56.51
(b) Effect of γ	
γ	Acc (%)
0.1	64.30
0.2	64.77
0.4	64.19
0.6	64.21
0.8	63.91
0.9	63.95
(c) Effect of τ	
τ	Acc (%)
0.05	62.03
0.10	63.75
0.15	64.77
0.20	64.42
0.25	63.84
0.30	63.68

sults confirm DNBR’s universal dual-dimensional bias correction mechanism: (1) SBB initially dilutes source-specific channel and spatial biases via parameter-free feature modulation, producing bias-diluted representations; (2) DNBR subsequently uses lightweight depthwise separable convolutions to generate target-adaptive channel importance weights and spatial preference weights, enabling precise feature calibration. The lightweight design ensures adaptability in few-shot scenarios while preventing overfitting.

The Influence of The κ , γ , and τ Values. In the ablation study of κ , γ , and τ , we first conduct one-factor ablation experiments (Table 7) to assess the impact of the sparse masking threshold κ and the gradient-weighting parameters γ and τ on model performance. The experimental results show that $\kappa = 0.4$, $\gamma = 0.2$, and $\tau = 0.15$ provide the best accuracy (64.77%) in the model. Specifically, smaller values of κ (e.g., 0.1 or 0.2) slightly decrease the performance, while larger values of κ (e.g., 0.8 and 0.9) cause a significant drop in performance. The optimal sparse masking threshold is therefore $\kappa = 0.4$. Similarly, for γ and τ , the best accuracy is achieved with $\gamma = 0.2$ and $\tau = 0.15$. Based on these findings, we further conduct a small-scale factorial study (Table 8) to

Table 8

Small-scale grid search ablation study on the effect of hyperparameters κ , γ , and τ (5-way 5-shot). Mean $\pm$ 95% CI over 5 seeds. We only present a subset of the best-performing experimental results from the small-scale grid search ablation study. 95% CI refers to 95% confidence intervals.

κ	γ	τ	Acc (%)	Mean Acc (%)	95% CI
0.3	0.1	0.10	63.59	64.01	± 0.36
0.3	0.2	0.10	64.03		
0.3	0.2	0.15	64.23		
0.3	0.3	0.15	64.10		
0.3	0.3	0.20	64.12		
0.4	0.1	0.15	64.30	64.23	± 0.30
0.4	0.2	0.10	63.75		
0.4	0.2	0.15	64.77		
0.4	0.2	0.20	64.42		
0.4	0.3	0.10	63.82		
0.4	0.3	0.20	64.33	64.12	± 0.44
0.5	0.1	0.20	64.16		
0.5	0.2	0.15	64.37		
0.5	0.2	0.20	64.21		
0.5	0.3	0.10	63.80		
0.5	0.3	0.15	64.04		

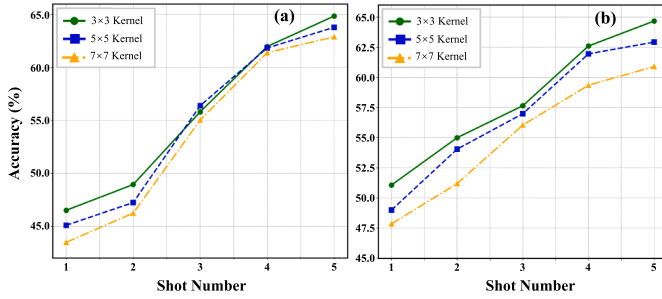


Fig. 8. Ablation study on the depthwise convolution kernel size in DNBR. (a) *Near-domain* and (b) *distant-domain* accuracy comparisons of 3×3 , 5×5 , and 7×7 kernels across different shot numbers. The 3×3 configuration consistently achieves superior performance, demonstrating its optimal balance between local pattern capture and noise suppression.

explore the interaction effects between κ , γ , and τ . By testing various hyperparameter combinations, the results confirm that the combination of $\kappa = 0.4$, $\gamma = 0.2$, and $\tau = 0.15$ delivers the optimal performance, with an accuracy of 64.77% and minimal fluctuation. These results demonstrate that this set of hyperparameters achieves good performance in few-shot cross-domain adaptation, further validating their superiority in the current task.

The Influence of Depthwise Convolution Kernel Size in DNBR. As shown in Fig. 8, our ablation studies on three kernel sizes (while maintaining feature map dimensions) demonstrate that 3×3 kernels achieve optimal performance. In 5-shot *distant-domain* tasks, 3×3 kernels yield 64.67% accuracy, significantly outperforming 5×5 (62.92%) and 7×7 (60.90%) configurations. This advantage persists in *near-domain* tasks (64.86% vs. 63.80%/62.90%), though with reduced performance gaps. The visualization results in Fig. 9 further confirm that 3×3 kernels' localized receptive field better captures transferable local patterns while effectively suppressing irrelevant global noise from source domains. Based on these comprehensive experiments, we adopt 3×3 as the default spatial mask configuration for all subsequent evaluations.

The Influence of MG-MAML. MG-MAML demonstrates consistent advantages over both standard MAML and random masking strategies through its innovative sparse gradient masking mechanism, as evidenced in Table 6 and Fig. 10. Our method achieves precise optimization of critical parameters while avoiding redundant updates of non-essential ones, resulting in superior performance across various experimental settings. Comparative results (rows 3-4, 6-7, 9-10, 11-12) show

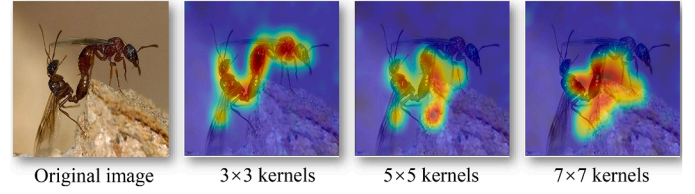


Fig. 9. Visualization of kernel size effects on feature extraction. Heatmaps demonstrate progressive pattern capture with increasing kernel sizes ($3 \times 3 \rightarrow 5 \times 5 \rightarrow 7 \times 7$). The 3×3 kernels maintain precise localization of anatomical features, while larger kernels exhibit broader but less discriminative activations.

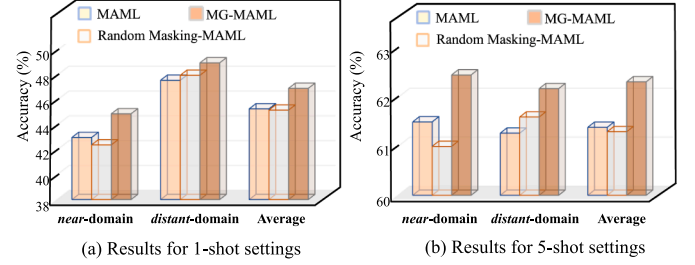


Fig. 10. Performance comparison under different MAML strategies: (a) 1-shot and (b) 5-shot classification accuracy across *near-domain* and *distant-domain* tasks. MG-MAML demonstrates consistent superiority over random masking and original MAML, validating its intelligent gradient-aware parameter selection. Random masking employed the same masking ratio as MG-MAML.

that MG-MAML maintains stable performance improvements while significantly reducing training instability, with a 43.44% decrease in inter-experiment variance compared to standard MAML. This advantage is especially evident in *distant-domain* tasks, due to its selective parameter freezing strategy that prevents source-domain overfitting while preserving feature generalization for target adaptation. The mechanism proves especially valuable in significant domain shifts, where traditional MAML's dense gradient updates tend to accumulate source biases, while MG-MAML's sparse constraints enable more stable learning of cross-domain shared representations. Fig. 10 further confirms MG-MAML's superiority over random masking, with average performance advantages of 1.75% and 1.02% in 1-shot and 5-shot tasks respectively, validating its intelligent parameter selection capability for effective domain adaptation.

The Influence of Channel/Spatial Mask. Fig. 11 presents classification results under 1-shot and 5-shot settings with individual and combined applications of channel and spatial masks. When used individually, channel masks outperform spatial masks, indicating the model's stronger dependence on source-domain channel patterns. The combined approach achieves 62.32% accuracy in a 5-shot setting, conclusively validating the necessity of dual-dimensional calibration. This synergistic effect proves particularly impactful in *distant-domain* scenarios (5.08% gain vs. 3.96% for *near-domain*), revealing that greater domain disparities necessitate simultaneous channel-spatial bias calibration. Channel masks mainly modulate feature responses, while spatial masks optimize attention distributions. Their synergistic interaction enables more comprehensive mitigation of source-domain biases. This finding provides critical guidance for feature calibration in cross-domain learning: single-dimensional bias calibration demonstrates limited effectiveness, necessitating the establishment of joint optimization mechanisms.

The Influence of NCSA. The ablation study of the NCSA module (Table 6, row 5) demonstrates its significant improvement in CD-FSL performance, achieving an average accuracy gain of 4.04% (4.68% for *distant-domain* tasks). By establishing bidirectional channel correlations and spatial semantic correspondences between support-query pairs, NCSA effectively enhances both discriminability and consistency

Table 9

Few-shot classification performance across different backbones. Top results are shown in **bold** (mean $\pm$ S.D.%), with all values representing percentages (%).

Methods	Backbones	Ave: near-domain		Ave: distant-domain		Average	
		1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
GNN + AFA [22]	Conv-4	30.35 $\pm$ 0.4	49.04 $\pm$ 0.3	40.18 $\pm$ 0.4	54.57 $\pm$ 0.4	35.27 $\pm$ 0.4	51.81 $\pm$ 0.3
	ResNet-10	42.98 $\pm$ 0.5	62.00 $\pm$ 0.4	46.49 $\pm$ 0.4	59.64 $\pm$ 0.4	44.74 $\pm$ 0.4	60.82 $\pm$ 0.4
LDP-net [23]	Conv-4	32.18 $\pm$ 0.4	50.06 $\pm$ 0.3	40.98 $\pm$ 0.5	55.69 $\pm$ 0.3	36.58 $\pm$ 0.5	52.88 $\pm$ 0.3
	ResNet-10	43.19 $\pm$ 0.3	61.16 $\pm$ 0.4	47.93 $\pm$ 0.4	61.54 $\pm$ 0.4	45.56 $\pm$ 0.3	61.35 $\pm$ 0.4
Ours	Conv-4	34.03 $\pm$ 0.4	52.41 $\pm$ 0.3	42.60 $\pm$ 0.3	57.81 $\pm$ 0.4	38.32 $\pm$ 0.4	55.11 $\pm$ 0.4
	ResNet-10	46.52 $\pm$ 0.4	64.86 $\pm$ 0.4	51.08 $\pm$ 0.3	64.67 $\pm$ 0.4	48.80 $\pm$ 0.4	64.77 $\pm$ 0.4

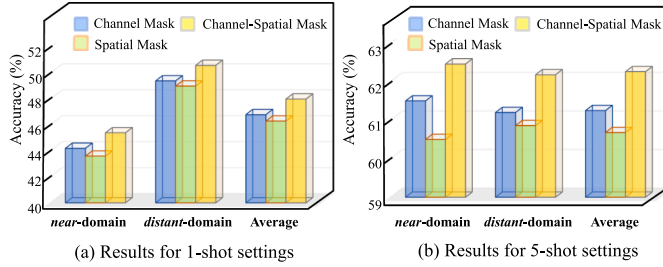


Fig. 11. Channel vs. Spatial vs. Channel-Spatial Mask Performance. (a) 1-shot and (b) 5-shot classification accuracy across domain configurations. Blue, green, and yellow bars indicate accuracy when using Channel, Spatial, and Channel-Spatial Mask separately.

of target-domain features. Experimental results reveal a positive correlation between the module's performance gains and domain discrepancy levels, with particularly notable improvements observed in *distant-domain* scenarios. Attention heatmap visualizations (Fig. 15) before and after applying NCSA clearly demonstrate its effectiveness in semantic alignment. These results collectively validate NCSA's capability in addressing cross-domain semantic alignment challenges under few-shot conditions.

Complexity Comparison. Our efficiency evaluation examines three key metrics: parameter scale (PS), computational cost (GFLOPs), and GPU computation duration (GCD) across 5-way 1-shot and 5-shot configurations. As shown in Table 10, compared with ProtoNet [59], GNN + AFA [22], LDP-Net [23], and GNN + FAP [60], CSDBC achieves improved performance while maintaining low resource overhead, making it suitable for real-time inference. Specifically, CSDBC has a parameter size of 8.18M, only slightly higher than ProtoNet (8.04M) by 0.14M, and significantly lower than GNN + AFA (9.47M), LDP-Net (9.11M), and GNN + FAP (11.30M). In terms of computational cost, CSDBC requires 101.726 GFLOPs and 127.311 GFLOPs for 1-shot and 5-shot tasks, respectively, which is very close to ProtoNet, with only a 0.17% and 0.29% increase, and far lower than other comparison methods. Regarding GCD, CSDBC takes only 0.00983 seconds for 1-shot tasks, faster than GNN + AFA, LDP-Net, and GNN + FAP; for 5-shot tasks, it takes 0.01530 seconds, also outperforming most comparison methods. Moreover, since CSDBC introduces task-dependent delays during the meta-adaptation process, we report the complete adaptation time and inference time of the model (as shown in Table 11) to quantify its efficiency in practical applications. In a single 5-way 1-shot or 5-shot task, the adaptation time and inference time together constitute the total prediction time, and the results show that our method can meet the requirements for real-time inference. In summary, our method achieves excellent performance while effectively controlling parameter size and computational cost, combining high efficiency with practicality and making it suitable for real-time deployment.

As shown in Table 6, the full model combining SBB, DNBR, and NCSA delivers the best results. Extensive ablation experiments confirm that each component contributes meaningfully to the overall perfor-

Table 10

Comparative analysis of model architectures, examining: parameter scale (PS), computational requirements (GFLOPs), and GPU computation duration (GCD) measured on NVIDIA 2080 Ti GPU.

Method	PS	5-way 1-shot		5-way 5-shot	
		GFLOPs	GCD	GFLOPs	GCD
ProtoNet [59]	8.04M	101.550	0.00960s	126.938	0.01300s
GNN + AFA [22]	9.47M	112.892	0.01120s	133.512	0.01842s
LDP-Net [23]	9.11M	109.703	0.01093s	129.007	0.01577s
GNN + FAP [60]	11.30M	127.103	0.01410s	141.023	0.01959s
Ours	8.18M	101.726	0.00983s	127.311	0.01530s

Table 11

Adaptation time and inference time of our method under different few-shot settings. All results are measured on NVIDIA 2080 Ti.

Method	Setting	Adaptation Time	Inference Time
Ours	5-way 1-shot	0.00829s	0.00154s
	5-way 5-shot	0.01353s	0.00177s

mance, while their joint integration further enhances generalization ability.

4.5. Numerical results: Comparison with state-of-the-art

To validate our method's effectiveness, we conduct comprehensive benchmarking of CSDBC against existing FSL methods. The evaluation encompasses both traditional FSL methods [8,46] and contemporary CD-FSL approaches that employ various adaptation strategies, including feature alignment [19], adversarial learning [22], and transformation techniques [6,23,54,57]. For rigorous comparative analysis, we organize the experimental results according to two distinct evaluation protocols: fine-tuning (FT) and transductive learning (TR) scenarios. This systematic categorization ensures fair comparison across different methodological paradigms while maintaining consistent evaluation standards.

Our evaluation employs the standard n -way k -shot protocol across eight diverse target domains, systematically categorized by domain shift magnitude: *near-domain* datasets (CUB [47], Cars [48], Places [49], Plantae [50]) and *distant-domain* benchmarks (ChestX [51], ISIC [52], EuroSAT [53], CropDisease [54]). The comprehensive results under varying domain shifts are detailed in Table 12.

To verify architectural compatibility, we conduct extensive evaluations using two established backbone networks: the lightweight Conv-4 and deeper ResNet-10 architectures [46,59]. As evidenced in Table 9, our method demonstrates robust performance across both feature extractors, with ResNet-10 showing particularly strong results. This deeper architecture's enhanced feature extraction capabilities prove especially suitable for our framework, given its emphasis on discriminative feature learning - a critical requirement for achieving high performance in fine-grained recognition scenarios. Consequently, we adopt ResNet-10 as our primary backbone for all subsequent experimental evaluations.

Table 12

Few-shot classification accuracy (%) trained with *mini*-ImageNet for 5-way 1-shot/5-shot tasks. **TR** (transductive learning) denotes methods utilizing full task data. Best performance shown in **bold** (mean $\pm$ S.D.%).

Model	TR	CUB		Cars		Places		Plantae	
		1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
MatchingNet [46] (2016)	$\times$	35.89 $\pm$ 0.5	51.37 $\pm$ 0.8	30.77 $\pm$ 0.5	38.99 $\pm$ 0.6	49.86 $\pm$ 0.8	63.16 $\pm$ 0.8	32.70 $\pm$ 0.6	46.53 $\pm$ 0.7
GNN [8] (2017)	$\times$	44.40 $\pm$ 0.5	62.87 $\pm$ 0.5	31.72 $\pm$ 0.4	43.70 $\pm$ 0.4	52.42 $\pm$ 0.5	70.91 $\pm$ 0.5	33.60 $\pm$ 0.4	48.51 $\pm$ 0.4
GNN + ATA [19] (2021)	$\times$	45.00 $\pm$ 0.5	66.22 $\pm$ 0.5	33.61 $\pm$ 0.4	49.14 $\pm$ 0.4	53.57 $\pm$ 0.5	75.48 $\pm$ 0.4	34.42 $\pm$ 0.4	52.69 $\pm$ 0.4
MN + AFA [22] (2022)	$\times$	41.02 $\pm$ 0.4	59.46 $\pm$ 0.4	33.52 $\pm$ 0.4	46.13 $\pm$ 0.4	54.66$\pm$0.5	68.87 $\pm$ 0.4	37.60 $\pm$ 0.4	52.43 $\pm$ 0.4
GNN + AFA [22] (2022)	$\times$	46.86 $\pm$ 0.5	68.25 $\pm$ 0.5	34.25 $\pm$ 0.4	49.28 $\pm$ 0.5	54.04 $\pm$ 0.6	76.21$\pm$0.5	36.76 $\pm$ 0.4	54.26 $\pm$ 0.4
LDP-net [23] (2023)	$\times$	49.82	70.39	35.51	52.84	53.82	72.90	39.84	58.49
GNN + FAP [60] (2024)	$\times$	46.91 $\pm$ 0.5	67.66 $\pm$ 0.5	34.69 $\pm$ 0.4	50.20 $\pm$ 0.4	54.41 $\pm$ 0.5	74.98 $\pm$ 0.4	36.73 $\pm$ 0.4	54.54 $\pm$ 0.4
FLoR [6] (2024)	$\times$	49.99	70.39	37.41	53.43	53.18	72.31	40.10	55.80
Ours	$\times$	51.03$\pm$0.4	71.70$\pm$0.4	39.22$\pm$0.3	54.80$\pm$0.5	54.40 $\pm$ 0.5	74.32 $\pm$ 0.4	41.41$\pm$0.3	58.60$\pm$0.4
TPN + AFA [22] (2022)	$\checkmark$	50.85 $\pm$ 0.4	65.86 $\pm$ 0.4	38.43 $\pm$ 0.4	47.89 $\pm$ 0.4	60.29 $\pm$ 0.5	72.81 $\pm$ 0.4	40.27 $\pm$ 0.4	55.67 $\pm$ 0.4
RDC [57] (2022)	$\checkmark$	48.68 $\pm$ 0.5	64.36 $\pm$ 0.4	38.26 $\pm$ 0.5	52.15 $\pm$ 0.4	59.53 $\pm$ 0.5	73.24 $\pm$ 0.4	42.29 $\pm$ 0.5	57.50 $\pm$ 0.4
CDCNet [61] (2025)	$\checkmark$	66.55 $\pm$ 0.8	84.67 $\pm$ 0.6	52.85 $\pm$ 0.8	70.08 $\pm$ 0.7	62.73 $\pm$ 0.8	80.77 $\pm$ 0.6	52.67 $\pm$ 0.8	71.65 $\pm$ 0.7
MCM [29] (2025)	$\checkmark$	67.68 $\pm$ 0.8	85.59 $\pm$ 0.6	55.43 $\pm$ 0.9	73.87 $\pm$ 0.7	63.52$\pm$0.9	81.77 $\pm$ 0.6	55.11 $\pm$ 0.9	73.62 $\pm$ 0.7
Ours	$\checkmark$	68.90$\pm$0.4	86.64$\pm$0.4	55.97$\pm$0.4	74.61$\pm$0.3	63.01 $\pm$ 0.3	81.95$\pm$0.4	56.76$\pm$0.4	75.49$\pm$0.3
		CropDiseases		EuroSAT		ISIC		ChestX	
		1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
MatchingNet [46] (2016)	$\times$	57.57 $\pm$ 0.5	73.26 $\pm$ 0.5	54.19 $\pm$ 0.5	67.50 $\pm$ 0.5	29.62 $\pm$ 0.3	32.98 $\pm$ 0.3	22.30 $\pm$ 0.2	22.85 $\pm$ 0.2
GNN [8] (2017)	$\times$	59.19 $\pm$ 0.5	83.12 $\pm$ 0.4	54.61 $\pm$ 0.5	78.69 $\pm$ 0.4	30.14 $\pm$ 0.3	42.54 $\pm$ 0.4	21.94 $\pm$ 0.2	23.87 $\pm$ 0.2
GNN + ATA [19] (2021)	$\times$	67.45 $\pm$ 0.5	90.59 $\pm$ 0.3	61.35 $\pm$ 0.5	83.75 $\pm$ 0.4	33.21 $\pm$ 0.4	44.91 $\pm$ 0.4	22.10 $\pm$ 0.2	24.32 $\pm$ 0.4
MN + AFA [22] (2022)	$\times$	60.71 $\pm$ 0.5	80.07 $\pm$ 0.4	61.28 $\pm$ 0.5	69.63 $\pm$ 0.5	32.32 $\pm$ 0.3	39.88 $\pm$ 0.3	22.11 $\pm$ 0.2	23.18 $\pm$ 0.2
GNN + AFA [22] (2022)	$\times$	67.61 $\pm$ 0.5	88.06 $\pm$ 0.3	63.12 $\pm$ 0.5	85.58$\pm$0.4	33.21 $\pm$ 0.3	46.01 $\pm$ 0.4	22.92 $\pm$ 0.2	25.02 $\pm$ 0.2
LDP-net [23] (2023)	$\times$	69.64	89.40	65.11	82.01	33.97	48.06	23.01	26.67
GNN + FAP [60] (2024)	$\times$	69.97 $\pm$ 0.5	91.79 $\pm$ 0.3	62.96 $\pm$ 0.5	82.52 $\pm$ 0.4	35.63 $\pm$ 0.4	47.60 $\pm$ 0.4	22.36 $\pm$ 0.2	25.31 $\pm$ 0.2
FLoR [6] (2024)	$\times$	73.64	91.25	62.90	80.87	38.11	51.44	23.11	26.70
Ours	$\times$	75.10$\pm$0.3	93.13$\pm$0.4	64.08 $\pm$ 0.3	82.97 $\pm$ 0.5	39.91$\pm$0.4	53.84$\pm$0.5	25.23$\pm$0.5	28.71$\pm$0.4
TPN + AFA [22] (2022)	$\checkmark$	72.44 $\pm$ 0.6	85.69 $\pm$ 0.4	66.17 $\pm$ 0.4	80.12 $\pm$ 0.4	34.25 $\pm$ 0.4	46.29 $\pm$ 0.3	21.69 $\pm$ 0.1	23.47 $\pm$ 0.2
RDC [57] (2022)	$\checkmark$	79.72 $\pm$ 0.5	88.90 $\pm$ 0.3	65.58 $\pm$ 0.5	77.15 $\pm$ 0.4	32.33 $\pm$ 0.3	41.28 $\pm$ 0.3	22.77 $\pm$ 0.2	25.91 $\pm$ 0.2
CDCNet [61] (2025)	$\checkmark$	—	—	—	—	—	—	—	—
MCM [29] (2025)	$\checkmark$	—	—	—	—	—	—	—	—
Ours	$\checkmark$	90.71$\pm$0.4	93.96$\pm$0.5	78.91$\pm$0.4	88.15$\pm$0.4	49.64$\pm$0.4	58.82$\pm$0.3	32.41$\pm$0.5	37.80$\pm$0.4

Near-Domain Dataset Results Analysis. Our CSDBC framework shows superior performance on benchmarks with limited domain shifts. Under the 5-shot setting, CSDBC achieves accuracy improvements of 1.31 %, 1.43 %, and 0.11 % on CUB [47], Cars [48], and Plantae [50] respectively, outperforming state-of-the-art methods including GNN + FAP [60] and FLoR [6]. It also outperforms conventional approaches such as MatchingNet [46] and GNN [8]. The outstanding performance on fine-grained datasets (CUB and Cars) highlights CSDBC's unique advantages in discriminative feature learning. This validates the framework's effectiveness in calibrating inherent biases - unlike FLoR [6] which relies solely on feature regularization, our dual-dimensional calibration mechanism more thoroughly mitigates the model's dependency on source-specific channel responses and spatial structures through target-adaptive channel-spatial soft masks generated by lightweight depthwise separable convolutions.

Notably, the slight performance decrease on Places compared to MN + AFA [22] confirms our design principle: when significant semantic overlap exists between source and target domains, SBBD's bias dilution mechanism may excessively suppress some shared features. This observation precisely demonstrates CSDBC's effective control over source-domain biases.

In the more challenging 1-shot setting, CSDBC maintains stable advantages with improvements of 1.04 %, 1.81 %, and 1.31 % on CUB, Cars, and Plantae respectively, verifying consistent performance across different sample sizes. Under transductive learning (TR) settings, CSDBC achieves optimal results across all test sets, demonstrating framework stability across evaluation protocols. These results confirm that systematic channel-spatial dual-dimensional bias calibration effectively addresses source-domain bias in CD-FSL. Our study primarily focuses on non-transductive settings as they better reflect real-world requirements

for model generalization. While we provide TR results in supplementary experiments, our main analysis concentrates on non-transductive performance.

Distant-Domain Dataset Results Analysis. The CSDBC framework demonstrates even more pronounced advantages on benchmarks with significant domain shifts. In typical *distant*-domain scenarios including medical imaging (ISIC) and agricultural analysis (CropDiseases), CSDBC achieves 1-shot accuracies of 39.91 % and 75.10 % respectively, outperforming the second-best methods by 1.80 % and 1.46 %. These improvements result from CSDBC's dual-dimensional bias calibration mechanism. First, the SBBD module effectively mitigates the model's over-reliance on source-specific channel responses and spatial structures through feature modulation. In medical imaging, SBBD suppresses the pretrained model's preference for color and texture features, enabling better adaptation to target-domain grayscale and local structures. Second, the DNBR and NCSA modules work in concert - generating target-adaptive channel-spatial soft masks and performing cross-image semantic alignment respectively. Compared to existing methods relying on simple feature enhancement, our approach more precisely focuses on discriminative target-domain features. This proves particularly crucial in specialized domains with significant shifts like medical imaging (2.40 % gain, 5-shot) and agricultural analysis (1.88 % gain, 5-shot).

Notably, CSDBC's average improvement on *distant*-domain datasets significantly exceeds that on *near*-domain scenarios, validating our core design principle: systematic channel-spatial dual-dimensional bias calibration more effectively addresses inherent bias problems caused by significant domain shifts. This advantage persists under extreme 1-shot conditions, demonstrating CSDBC's practical value for real-world applications like medical diagnosis and agricultural monitoring.

Table 13

Few-shot classification accuracy (%) trained with *mini*-ImageNet for 5-way 1-shot/5-shot tasks. FT indicates fine-tuning on target domain and TR denotes full task data utilization. Best performance shown in **bold** (mean $\pm$ S.D.%).

Model	FT	TR	CUB		Cars		Places		Plantae	
			1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
Fine-tuning [5] (2020)	✓	✗	43.53 $\pm$ 0.4	63.76 $\pm$ 0.4	35.12 $\pm$ 0.4	51.21 $\pm$ 0.4	50.57 $\pm$ 0.4	70.68 $\pm$ 0.4	38.77 $\pm$ 0.4	56.45 $\pm$ 0.4
NSAE [20] (2021)	✓	✗	–	68.51 $\pm$ 0.8	–	59.41 $\pm$ 0.7	–	71.02 $\pm$ 0.7	–	59.55 $\pm$ 0.7
FLoR [6] (2024)	✓	✗	50.01	73.39	38.13	57.21	53.61	72.37	40.20	61.11
Ours	✓	✗	50.72$\pm$0.4	74.10$\pm$0.3	39.23$\pm$0.3	58.91$\pm$0.3	54.80$\pm$0.3	73.88$\pm$0.5	41.10$\pm$0.4	63.12$\pm$0.4
TPN + ATA [19] (2021)	✓	✓	51.89 $\pm$ 0.5	70.14 $\pm$ 0.4	38.07 $\pm$ 0.4	55.23 $\pm$ 0.4	57.26 $\pm$ 0.5	73.87 $\pm$ 0.4	40.75 $\pm$ 0.4	59.02 $\pm$ 0.4
RDC [57] (2022)	✓	✓	51.20 $\pm$ 0.5	67.77 $\pm$ 0.4	39.13 $\pm$ 0.5	53.75 $\pm$ 0.5	61.50 $\pm$ 0.6	74.65 $\pm$ 0.4	44.33$\pm$0.6	60.63 $\pm$ 0.4
FLoR [6] (2024)	✓	✓	55.94	74.06	40.01	57.98	61.27	74.25	41.70	61.70
Ours	✓	✓	56.55$\pm$0.4	75.40$\pm$0.3	41.31$\pm$0.4	59.09$\pm$0.4	62.50$\pm$0.4	75.90$\pm$0.5	43.42 $\pm$ 0.4	63.31$\pm$0.4
			CropDiseases		EuroSAT		ISIC		ChestX	
			1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
Fine-tuning [5] (2020)	✓	✗	73.43 $\pm$ 0.5	89.84 $\pm$ 0.3	66.17 $\pm$ 0.5	81.59 $\pm$ 0.3	34.60 $\pm$ 0.3	49.51 $\pm$ 0.3	22.13 $\pm$ 0.2	25.37 $\pm$ 0.2
NSAE [20] (2021)	✓	✗	–	93.14 $\pm$ 0.5	–	83.96 $\pm$ 0.6	–	54.04 $\pm$ 0.6	–	27.10 $\pm$ 0.4
FLoR [6] (2024)	✓	✗	84.04	92.33	69.13	83.06	38.81	56.74	23.12	26.77
Ours	✓	✗	84.95$\pm$0.4	93.62$\pm$0.4	71.15$\pm$0.5	85.52$\pm$0.4	40.14$\pm$0.4	58.53$\pm$0.5	24.80$\pm$0.3	28.89$\pm$0.3
TPN + ATA [19] (2021)	✓	✓	82.47 $\pm$ 0.5	93.56 $\pm$ 0.2	70.84 $\pm$ 0.5	85.47 $\pm$ 0.3	35.55 $\pm$ 0.4	49.83 $\pm$ 0.3	22.45 $\pm$ 0.2	24.74 $\pm$ 0.2
RDC [57] (2022)	✓	✓	86.33 $\pm$ 0.5	93.55 $\pm$ 0.3	71.57 $\pm$ 0.5	84.67 $\pm$ 0.3	35.84 $\pm$ 0.4	49.06 $\pm$ 0.3	22.27 $\pm$ 0.2	25.48 $\pm$ 0.2
FLoR [6] (2024)	✓	✓	86.30	93.60	71.38	83.76	41.67	57.54	23.12	26.89
Ours	✓	✓	87.42$\pm$0.5	94.90$\pm$0.5	72.31$\pm$0.3	85.73$\pm$0.5	42.74$\pm$0.3	59.24$\pm$0.4	24.94$\pm$0.3	28.80$\pm$0.3

4.6. Numerical results: Comparison with fine-tuning

Prior research has established that fine-tuning pretrained models frequently surpass traditional FSL techniques in cross-domain scenarios [5,12]. Our experimental design expands this comparison beyond standard FSL and CD-FSL approaches to incorporate cutting-edge fine-tuning methods. All evaluated techniques share the same ResNet-10 backbone for feature extraction to ensure equitable comparison. Where fine-tuning methods employ conventional fully-connected classifiers, our CSDBC framework introduces specialized architectural enhancements to the base network. Adhering to the evaluation methodology from [19], we structure target tasks using the standard n -way k -shot paradigm with $q=15$ query instances per category. For a fair comparison with fine-tuning baselines, we apply the augmentation strategy from [57] to generate 15 synthetic samples per support class during model adaptation. Both CSDBC and fine-tuning baselines utilize identical training data under 5-way 1-shot/5-shot settings, trained for 50 epochs with consistent optimization settings: SGD ($\text{lr}=0.01$, momentum=0.9) for fine-tuning versus Adam ($\text{lr}=0.001$) for our approach. Table 13 demonstrates CSDBC's performance advantages, particularly pronounced on *distant*-domain benchmarks. In 5-shot evaluations, our method outperforms the second-best fine-tuning approach by 2.12% (ChestX), 1.79% (ISIC), 2.46% (EuroSAT), and 1.29% (CropDisease) - margins significantly wider than those observed on *near*-domain datasets (0.71%, 1.70%, 1.51%, and 2.01% respectively, detailed in Table 13). This performance gap stems from CSDBC's dual-component architecture: its bias dilution module reduces misleading inductive biases, while its feature recombination mechanism enhances discriminative learning. These advantages persist consistently across transductive learning (TR) and 1-shot configurations, aligning with the findings in Section 4.5.

4.7. Visualization of key results

To provide intuitive insights into our method's effectiveness, we perform visual analyses of critical components and their impacts.

Visualization of Static Base-class Bias Dilution. Fig. 12 presents comparative visualizations of feature transformations across both *near*-domain and *distant*-domain scenarios, employing box plots and line charts to contrast the distributions before (red) and after (blue) static base-class bias dilution. The visual analysis yields several significant

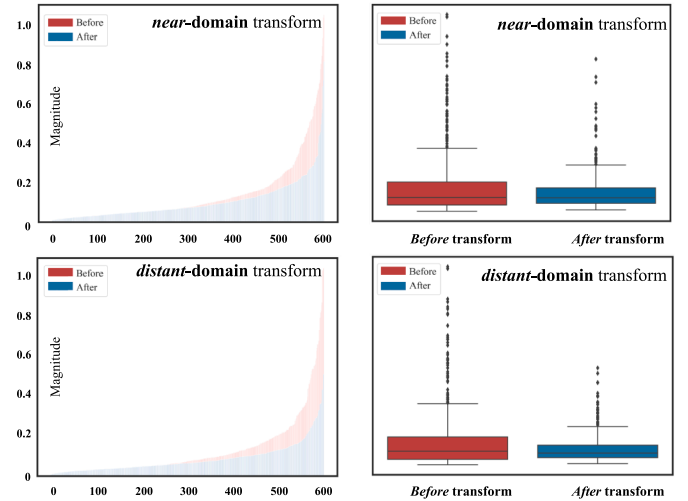


Fig. 12. Visualization of feature magnitude transformation before and after base-class bias dilution. Upper part: *near*-domain dataset transformation, line chart (left part) showing the trend of feature magnitude changes before (red) and after (blue) bias dilution, and boxplot (right part) comparing overall feature distributions. Lower part: corresponding visualizations for the *distant*-domain dataset. This line-box collaborative visualization reveals how feature distributions change during base-class bias dilution.

findings. Transformed features demonstrate substantially condensed value ranges with tighter distributions, manifested through reduced interquartile spans and diminished outlier presence. The bias dilution mechanism achieves more equitable spatial attention distribution, shifting focus from concentrated areas to broader regions. This spatial reorganization empowers the CSDBC framework to aggregate comprehensive visual information while reducing dependence on extreme feature values. For *distant*-domain samples particularly, the static base-class bias dilution effect enables superior extraction of discriminative information from spatially dispersed relevant regions, counteracting the inherent tendency for domain-specific feature clustering. The visualization results empirically validate that our approach generates more homogeneous feature distributions, with the transformation effects being especially pronounced in cross-domain scenarios. Comparative plots clearly

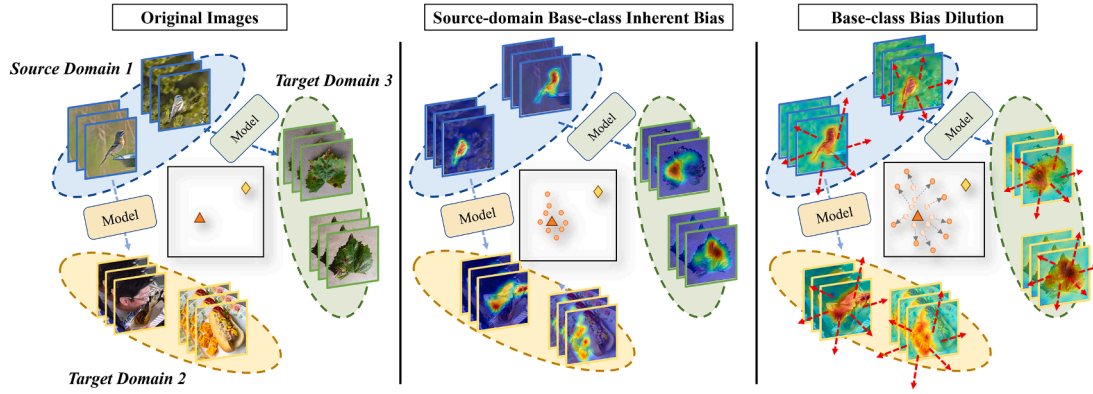


Fig. 13. Attention heatmaps before and after base-class bias dilution. Source domain: CUB; target domains: *mini*-ImageNet and CropDiseases. The second and third columns show the attention heatmaps before and after base-class bias dilution, respectively. In the second column, source-domain samples maintain precise discriminative region focus, while target-domain samples initially localize to incorrect features due to source-domain bias. In the third column, these erroneous source biases are diluted. The red arrows indicate the direction of bias dilution.

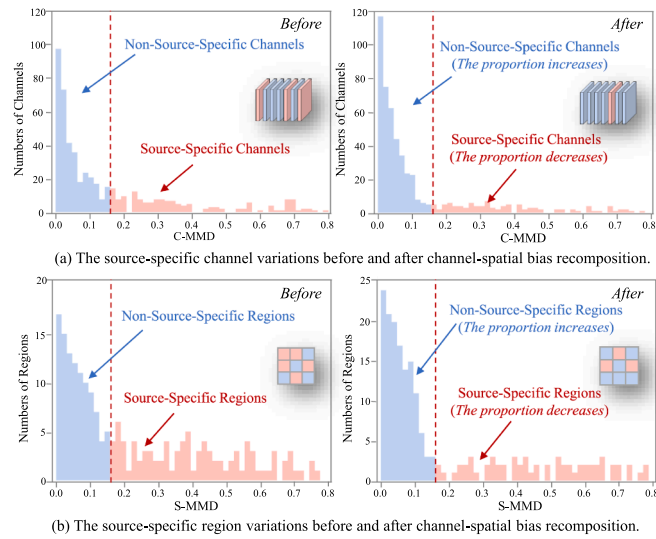


Fig. 14. Visualization of channel-spatial bias recomposition effects. (a) The source-specific channel variations before and after channel-spatial bias recomposition; (b) the source-specific region variations before and after channel-spatial bias recomposition. (Left) Original feature distributions exhibit pronounced source-specific patterns in both channel (C-MMD) and spatial (S-MMD) dimensions. (Right) Post-DNBR processing shows a significant reduction in both source-specific channels and regions.

show how CSDBC produces features with better-balanced statistical and spatial properties versus original distributions.

Heatmap Comparison: Before and After Bias Dilution. We analyze attention distribution changes using comparative heatmaps with CUB as source domain and *mini*-ImageNet/CropDisease as target domains. Fig. 13 reveals three fundamental findings: (1) the second column verifies appropriate attention localization in source-domain samples; (2) uncorrected models exhibit misguided focus in target domains, frequently attending to irrelevant regions due to base-class biases; and (3) after SBB processing (the third column), attention patterns become more spatially balanced with diminished emphasis on incorrect areas. This transformation validates CSDBC's capability via static base-class bias dilution to effectively redistribute attention weights across image regions, successfully overcoming the feature selection biases inherited from source domain training. Comparative visualizations show how our method achieves better spatial attention allocation while suppressing misleading focus patterns.

Visualization of Channel-Spatial Bias Recomposition. Fig. 14 shows that DNBR processing effectively reduces the model's reliance

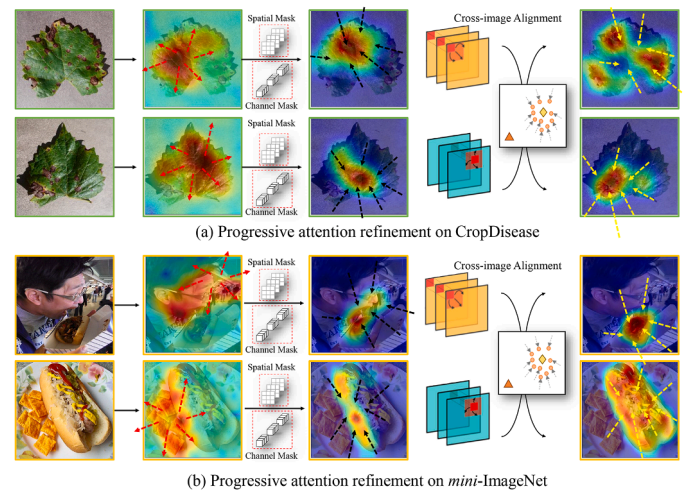


Fig. 15. Progressive attention refinement through bias dilution, recomposition, and alignment. Each subfigure consists of four columns: the first column shows original input images; the second column displays bias-diluted attention heatmaps with suppressed source-specific patterns; the third column presents bias-recomposed heatmaps; the final column demonstrates cross-image aligned heatmaps. Bias dilution (red arrows), bias recomposition (black arrows), and bias alignment (yellow arrows).

on source-specific patterns, as evidenced by changes in C-MMD and S-MMD distributions. In the channel dimension, the originally abundant source-specific channels are significantly reduced. Several source-specific channels that previously performed well in the source domain but exhibited significant performance degradation in the target domain were transformed into non-source-specific channels after bias recomposition. In the spatial dimension, DNBR significantly enhances the model's ability to extract spatial features from the target domain. Through analysis of the changes in S-MMD distributions, it is evident that the originally prevalent source-specific regions—those excessively activated in the source domain but lacking discriminative power in the target domain—are effectively suppressed. These regions typically correspond to source-specific textures or structures that would misdirect the model's attention to irrelevant areas in cross-domain scenarios.

Heatmap Comparison: Before and After Cross-image Alignment. Fig. 15 presents the complete transformation process of four target-domain samples, tracing the attention evolution from initial bias dilution through bias recomposition to cross-image semantic alignment. Each sequence shows: (1) second-column heatmaps revealing atten-

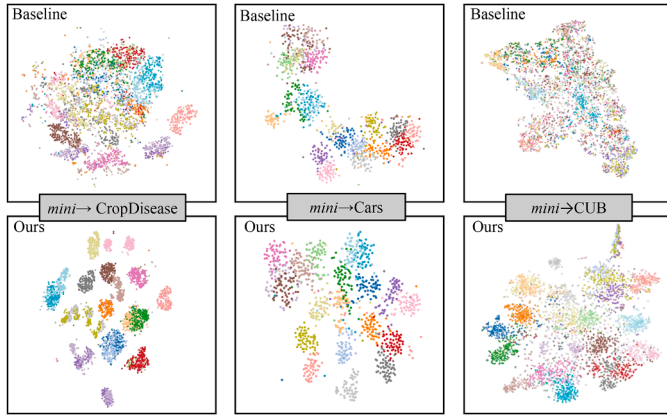


Fig. 16. t-SNE visualization comparing feature distributions between Baseline and Ours on three cross-domain datasets: *mini-Imagenet*→*CropDisease*, *mini-Imagenet*→*Cars*, *mini-Imagenet*→*CUB*. The baseline exhibits scattered clusters with inter-class overlap, while our method demonstrates tighter intra-class grouping and clearer inter-class separation.

tion distribution after bias dilution; (2) third-column heatmaps showing restructuring after dual-dimensional bias recombination; and (3) final-column heatmaps displaying optimized attention with precise feature localization through cross-image alignment. The NCSA module successfully enables the accurate extraction of cross-image discriminative features through this process. The visualization confirms our method's effectiveness in simultaneously suppressing inherent source-domain biases while discovering discriminative fine-grained features, ultimately enhancing model reliability. This progressive transformation from bias dilution to adaptive recombination and finally cross alignment clearly illustrates the framework's capability to eliminate misleading source-specific patterns while developing robust cross-domain representations.

t-SNE Visualization. We analyze feature distributions using t-SNE visualization across three cross-domain transfers: *mini*→*CropDisease*, *mini*→*Cars*, and *mini*→*CUB*. Fig. 16 contrasts the baseline (top row) with our method (bottom row). The baseline suffers from substantial inter-class mixing and ambiguous cluster boundaries, indicating poor domain adaptation. Conversely, our approach produces well-separated clusters with tight intra-class grouping across all datasets, demonstrating effective learning of discriminative target-domain features.

5. Conclusion

In this work, we present CSDBC, a channel-spatial dual-dimensional bias calibration framework that systematically addresses inherent source-domain inductive biases in CD-FSL through coordinated bias dilution and target-adaptive recombination. Our framework introduces three key innovations: (1) the SBB module pioneers parameter-free feature modulation to dilute channel-wise response dependencies and spatial structural preferences from the source domain without compromising representational capacity; (2) the DNBR module leverages meta-optimized depthwise separable convolutions with our novel MG-MAML strategy to dynamically recombine target-domain-adaptive channel-spatial biases; and (3) the NCSA module establishes bidirectional cross-image correlations, enhancing feature discriminability through channel-spatial semantic alignment. Extensive experiments across eight benchmarks demonstrate that CSDBC consistently outperforms existing methods, validating its effectiveness in tackling dual-dimensional biases. This work provides significant theoretical and practical contributions to CD-FSL as the first to formally identify and address channel-spatial dual biases. The proposed bias dilution-recomposition paradigm opens new directions for inductive bias calibration, potentially inspiring more

generalizable FSL frameworks. Future research may extend our dual-dimensional calibration principles to other cross-domain adaptation scenarios, such as few-shot object detection or segmentation, where channel and spatial biases similarly impede generalization.

CRedit authorship contribution statement

Minghui Li: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization; **Hongxun Yao:** Writing – review & editing, Validation, Supervision, Data curation.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hongxun Yao reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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